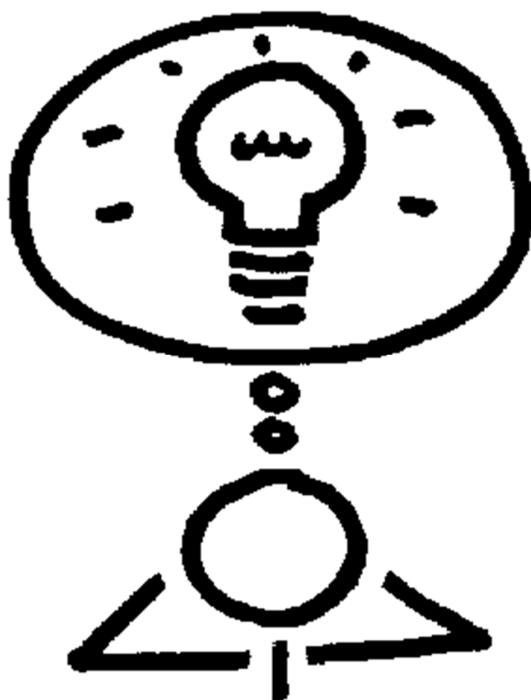


Physics Worksheets



Year 9

Compiled by Rodney Bugeja B.Ed (Hons)

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Density

1. Explain an experiment (using diagrams) to find the density of:

a) A cube of aluminium



.....

[3]

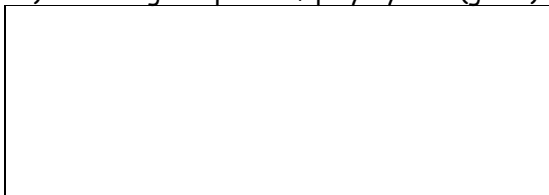
b) An irregular shaped piece of rock



.....

[3]

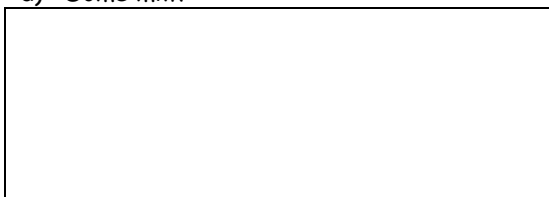
c) An irregular piece of polystyrene (gablo)



.....

[3]

d) Some milk



.....

[3]

2. A stone of mass of 30g is placed in a measuring cylinder containing some water. The reading of the water level increases from 50cm^3 to 60cm^3 . What is the density of the water?

.....
[1]

3. An object has a mass of 40000kg and a volume of 5m^3 .

a) What is its density?

.....[1]

b) What material do you think it could be?

.....[1]

4. The density of air is 1.3kg/m^3 .

a) What mass of air is contained in a room measuring $2.5\text{m} \times 3.5\text{m} \times 11\text{m}$?

.....
[1]

b) How does this compare to your mass?

.....[1]

5. Gorg wants to load some bricks into his van. There are 1000 bricks, and when stacked neatly they measure 2m by 1m by 1m.

a) What is the volume of the stack?

.....[1]

b) What is the volume of one brick?

.....[1]

c) If the density of brick is 2500kg/m^3 , what is the mass of the stack?

.....
.....[2]

d) If his van's maximum load is 1000kg, how many bricks can he load?

.....[2]

6. An iron rod has a density of 8g/cm^3 . Its mass is 10kg.

a) Find its volume in cm^3

.....[1]

b) Find its volume in m^3

.....[1]

7. King Henry ordered his goldsmith to make him a new crown, but when it was finished he was suspicious that it was only plated with gold and not solid.

a) Can you help King Henry to devise an experiment to test if the crown is solid gold or not?



.....
.....
.....
.....[3]

b) From your experiment he found its mass was 1125g and its volume is 114cm^3 . What do you think?

.....[1]

8. Complete the table below:

Object	Density (kg/m^3)	Mass (kg)	Volume (m^3)
A		4000	2
B	8000		4
C	2000	1000	
D		2000	4

.....
.....
.....
.....[4]

a) Which objects could be made of the same material? Why?

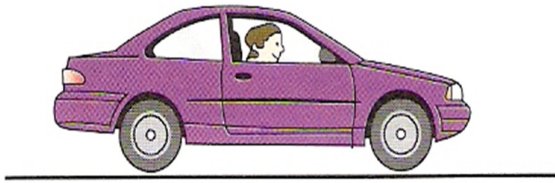
.....[1]

b) Which object would float on water? Why?

.....[1]

Forces

1. The diagram shows a car moving from left to right:



- a) On the diagram use arrows to show:
- The weight of the car
 - The reaction force of the ground on the car
 - The driving force
 - The air resistance
 - The friction

[5]

- b) Describe the motion of the car if:

- i) The driving force and the resistive forces (friction & air resistance) are equal

.....[1]

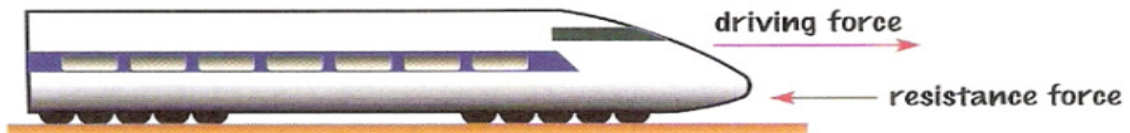
- ii) The driving force is greater than the resistive forces

.....[1]

- iii) The driving force is greater than the resistive forces

.....[1]

2. A train is moving along a track. Describe the forces acting on the train when:



- a) The train accelerates

.....[1]

- b) The train moves with a constant speed

.....[1]

- c) The train slows down (decelerates)

.....[1]

3. The diagrams show the forces on 4 different objects. They all start from rest. For each one:

- Calculate the resultant force
- State whether it will move left, right or remain stationary



.....[2]



.....[2]



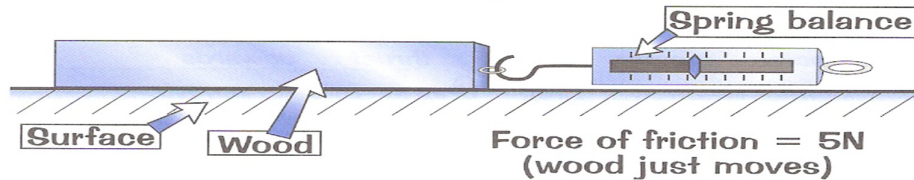
.....[2]



.....[2]

Friction & Weight

1. The diagram shows how the force of friction can be measured:



a) In which direction is friction acting?
.....[1]

b) Give two ways you can increase the force of friction
.....[1]

c) Give three ways of decreasing friction
.....[1]

2. Riding a bicycle gives us plenty of examples of friction being useful and a nuisance. Divide these up into "useful" and "nuisance" and explain each answer

Example:	Useful or nuisance?	Explanation
Air resistance		
Saddle		
Handlebar		
Pedals		
Wheel bearings		
Tyres		
brakes		

3. "Mass and weight" are used in everyday language almost as if they were the same thing. Decide which of the following is mass or weight:

a) Amount of matter		b) Is a force	
c) Measured in newtons		d) Caused by pull of gravity	
e) Same anywhere in the Universe		f) Is lower on the moon than on Earth	
g) Not a force		h) Measured in kg	
i) Measured by spring balance		j) Measured by a balance	

4. The strength of gravity on Earth is 10N/kg and the strength of gravity on the Moon is 1.6N/kg

a) Find the weight of rocks on Earth and the moon with the following masses:

i) 5kg ii) 11kg iii) 2.5kg.....
.....[1][1][1]

b) Find the mass of rocks with the following weights on Earth

i) 30N ii) 143N..... iii) 111.2N.....
.....[1][1][1]

5. An object has a mass of 4 kg.

(a) What is its weight here on Earth?
.....[1]

(b) What is his weight on the moon? (gravity on the moon = 1.67N/kg)
.....[1]

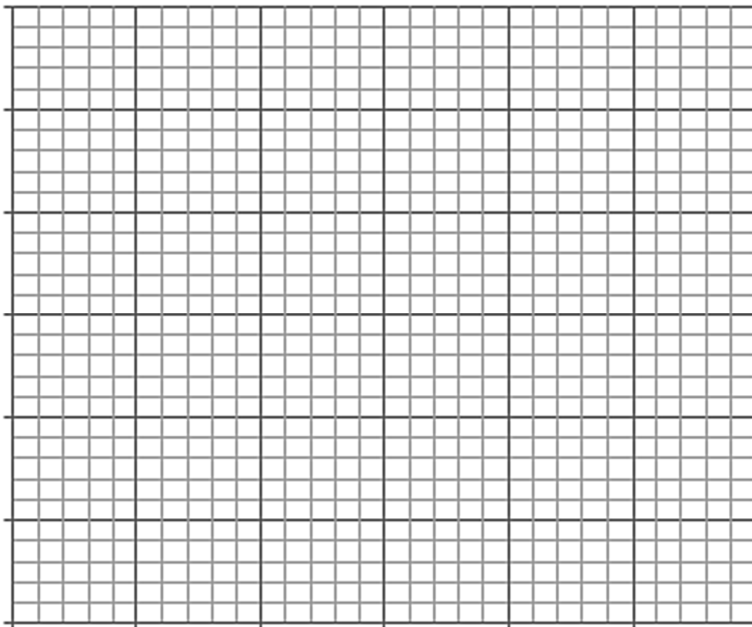
(c) What is his mass on the moon?
.....[1]

Hooke's Law

1. John carried out an experiment to investigate how the stretching force affects the extension of a spring. The original length of the spring was 2cm. The table of results obtained was:

Load (N)	0	1	2	3	4	5	6	7	8	9
Length (cm)	2	3.1	4.0	4.9	6.0	7.0	8.3	9.8	11.6	15.0
Extension (cm)										

- a) Complete the table [2]
b) Plot a graph of extension against load [3]



- c) On your graph label the elastic limit [1]
d) What conclusion can you draw from this experiment?
.....
.....
.....[2]
e) From your graph find:
i) The load that gives an extension of 7.5cm [1]
ii) The extension when the load is 2.5N..... [1]
iii) The length of the spring when the load is 3.4N..... [1]

2. A spring is 30cm long when it carries a load of 10N and 60cm long when it carries a load of 30N.

Work out the following:

- a) The length of the spring with no load[1]
b) The length of the spring when it has a load of 40N[1]
c) The extension of the spring when it has a load of 55N[1]
d) The load required to produce an extension of 45cm[1]
e) The load required to produce a spring of length 105cm[1]

3. A wire has an unstretched length of 60cm. When it carries a load of 10N its length becomes 61cm. The elastic limit is reached with a load of 50N

- a) Estimate the length of the wire when the load is:

i) 60N[1] ii) 70N.....[1] iii) 100N.....[1]

- b) What would eventually happen to the wire if we continually increase the load?

.....[1]

Newton's Laws of Motion

1. Use the words below to fill in the blanks

<i>proportional</i>	<i>force</i>	<i>reaction</i>	<i>stationary</i>	<i>accelerates</i>
<i>opposite</i>	<i>constant</i>	<i>resultant</i>	<i>inversely</i>	<i>Balanced</i>
	<i>Second</i>	<i>Third</i>	<i>First</i>	

Newton's Law: For every action there is an equal and

Newton's Law: If the forces on an object are, it's either or moving at velocity

Newton's Law: If an object has a force acting on it, it in the direction of the The acceleration is to the force and to the mass

2. Newton's First Law of Motion states that **balanced forces** means no change in **velocity**.

- a) Explain clearly the underlined terms

.....
[1]

- b) Draw a diagram of a car moving at constant velocity. Draw ALL horizontal forces. [2]

- c) Describe what is meant by **RESULTANT FORCE**

.....[1]

- d) For your diagram in (b) what is the resultant force?

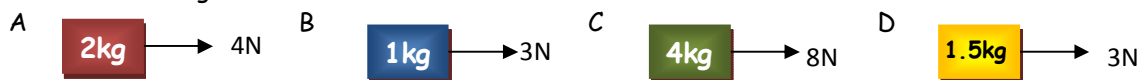
.....[1]

3. Newton's Second Law of Motion states that a non-zero resultant force means acceleration. Complete:

<i>Force</i>	<i>unequal</i>	<i>slowing down</i>	<i>speeding up</i>	<i>stopping</i>
<i>accelerate</i>	<i>decelerate</i>	<i>direction</i>	<i>starting</i>	

If there is an unbalanced, then an object will or in that this change in motion can take five different forms:,, and changing direction. On a force diagram, the arrows will be

4. Look at the four diagrams below:

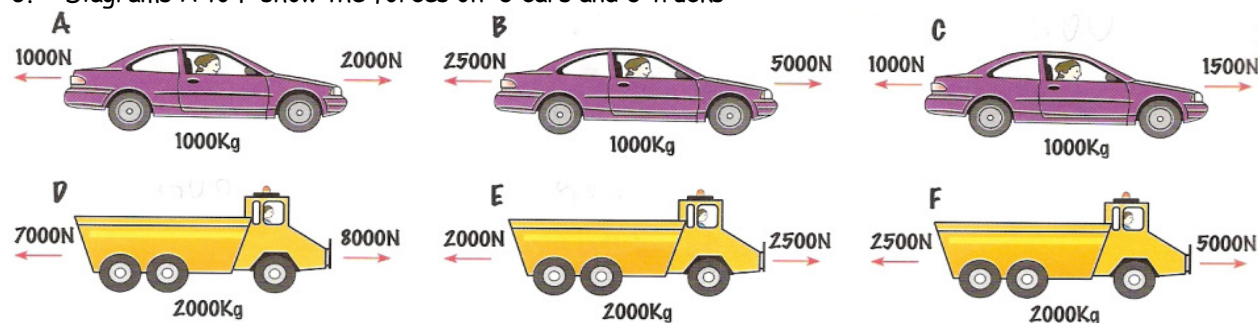


Which have the same acceleration? (show ALL working)

.....
[1]

Force, Mass & Acceleration

6. Diagrams A to F show the forces on 3 cars and 3 trucks



a) Which car has the smallest acceleration? Explain.

.....[1]

b) Which truck has the greatest acceleration? Explain.

.....[1]

c) Will car B or truck F have the greatest acceleration? Explain.

.....[1]

d) Which two vehicles have the same acceleration? Explain.

.....[1]

7. A car is moving at **constant velocity**, producing a driving force of 5000N.

a) What is the value of the resistive forces?

.....[1]

b) The driver accelerates and increases the driving force to 7000N

i) What is the resultant force now? (assume that the resistive force remains the same)

.....[1]

ii) Calculate the acceleration of the car if the mass is 800kg

.....[1]

8. A car of mass 1000kg is moving. The total resistive force is 700N. The **resultant force** is 500N.

a) Calculate the driving force

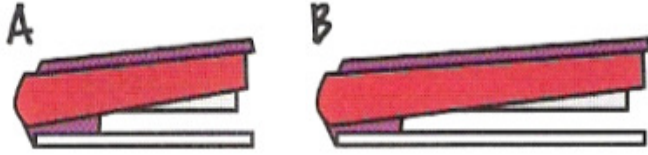
.....[1]

b) Calculate the acceleration

.....[1]

Moments

1. Explain the following:



a) Which stapler is easy to use? Explain

.....

[2]

b) Door handles are placed a long way from the hinges?

.....[1]

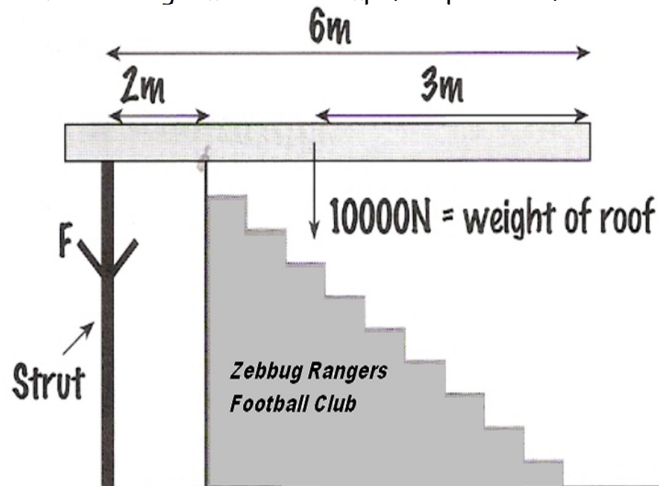
2. A mechanic applies a force of 200N at the end of a spanner of length 15cm. What moment does the spanner exert?

.....[1]

3. The diagrams show 6 rulers. For each one calculate the value of X

(i) [1]	(ii) [1]	(iii) [1]
(iv) [1]	(v) [1]	(vi) [1]

4. The diagram shows a simplified picture of the roof of a stand of a football ground.



a) Calculate the moment due to the weight of the roof of the stand

.....

[1]

b) The strut has to exert a force on the roof of the stand to balance it. Calculate this force

.....

[2]

Centre of Gravity

1. Explain the meaning of Centre of Gravity (mass)

.....
[1]

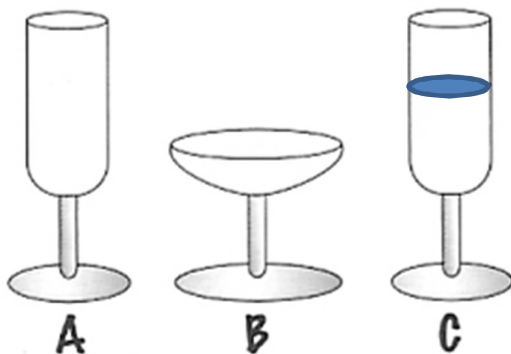
2. Describe an experiment to find the centre of gravity of an irregular flat card (lamina). Also draw a diagram.



.....

[3]

3. The diagram shows 3 different wine glasses:



a) Which is the most stable? Explain

.....

[2]

b) Which is the least stable? Explain

.....

[2]

4. Explain the following:

a) A Bunsen burner has a wide base

.....
[1]

b) A 4x4 truck has a wide wheel base

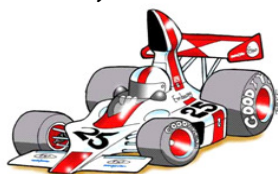
.....
[1]

c) A car becomes more unstable with a box on the roof

.....
[1]



d) Formula 1 cars are low with a wide base



.....
[1]

Pressure in Solids

1. Explain the following:

a) A 4x4 vehicle has wide tyres

.....[1]

b) Woodpeckers have strong pointed beaks

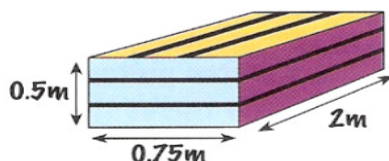
.....[1]

c) Camels have very large feet

.....[1]



2. The crate shown has a mass of 60kg.



a) Calculate the least pressure the crate can exert on the ground

.....

.....[2]

b) In what position will the crate exert a pressure of 1600Pa on the ground? Show all calculations.

.....

.....

.....[2]

3. A car weighs 10000N. Each tyre has an area of 100cm^2 in contact with the ground.

a) What is the total area in m^2 of the 4 tyres in contact with the ground?

.....

.....

.....[2]

b) Calculate the pressure exerted on the ground by the car.

.....

.....

.....[2]

4. A lorry has a mass of 32,000kg. Each tyre has an area of 400cm^2 in contact with the ground.

a) Calculate the pressure exerted on the ground if the lorry has 6 tyres.

.....

.....

.....[2]

b) Another lorry has a weight of 420,000N. Each tyre has an area of 400cm^2 . Calculate the number of tyres the lorry has if it exerts a pressure of 105N/cm^2 on the ground.

.....

.....

.....[2]

Pressure in liquids & Hydraulics

1. Explain:

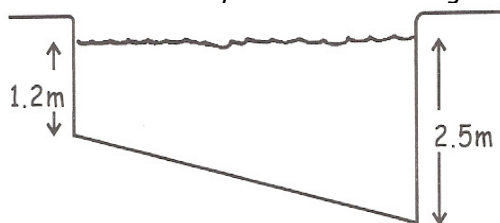
a) Why deep sea divers need to wear strong wet suits

.....[1]

b) The strength of the ship has to be greater the further it is under water

.....[1]

2. The density of water is 1000kg/m^3 . Calculate the pressure at the bottom of the swimming pool...



a) At the shallow end

.....[2]

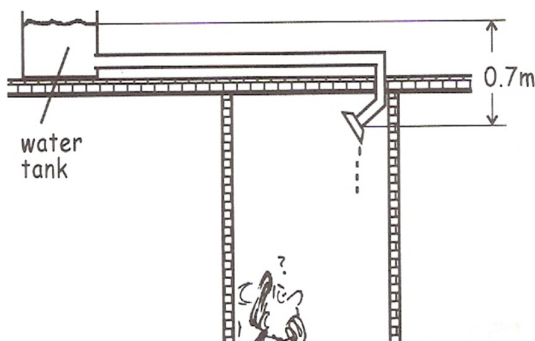
b) At the deep end

.....[2]

3. Paula's shower is not working properly because the pressure is too low.

a) How can she solve the problem?

.....[1]



a) What is the water pressure at the shower end?

.....[2]

b) To what height must the tank be raised so that the water pressure at the shower end is 15000Pa ?

.....[2]

4. A driver pushes on a brake pedal with 200N and the area of the master piston is 2cm^2 .

a) Calculate the pressure at the master piston.

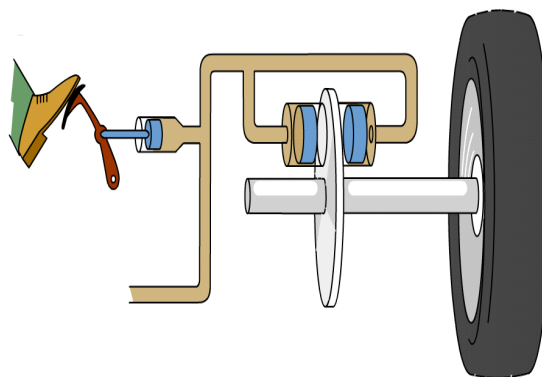
.....[2]

b) What is the pressure at the slave piston?

.....[2]

c) If the area of the slave piston is 16cm^2 , find the pressure at the slave piston

.....[2]



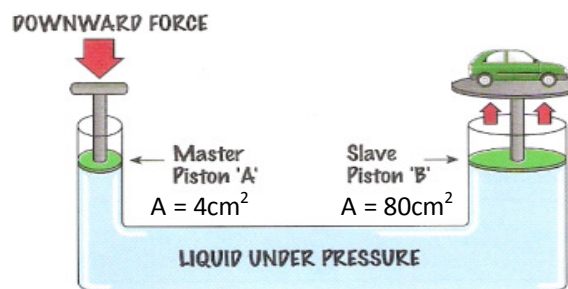
5. The diagram shows a hydraulic jack

a) Explain how it works

.....[2]

b) If the weight of the car is 10000N calculate the force to lift the car.

.....[2]



Pressure in Gases

1. Explain what we mean by 'atmospheric pressure'

.....
[1]

2. Why do aeroplanes need pressurized cabins?

.....
[1]

3. John carried out an experiment to investigate how the volume of a fixed mass of gas changes with pressure. He kept the temperature constant throughout the experiment.

- a) Complete the table:

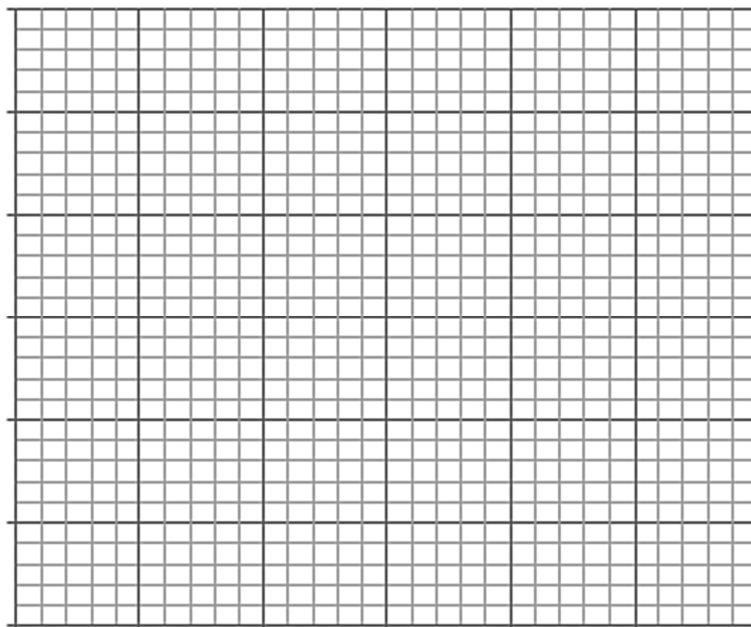
Pressure (atm)	Volume (cm ³)	Pressure x Volume
1	10.0	10
2	5.0	
3	3.3	
4	2.5	
5	2.0	

- b) Plot a graph of Pressure vs. Volume
 c) What conclusion can you get from the experiment?

.....[1]

- d) A gas cylinder stores gas at a pressure of 10atm. If the volume of the gas is 1000cm³ calculate the volume the gas will occupy in the atmosphere

.....
[1]



4. A weather balloon is released into the atmosphere. At ground level the volume of air in the balloon is 400cm³. The pressure is 100000Pa.

- a) If the balloon rises into the air to a height where the pressure is 25000Pa (assuming the temperature does not change), what is the new volume of the balloon?

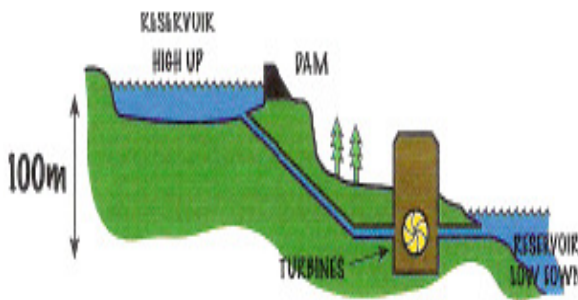
.....
[2]

- b) The balloon now changes its position and volume is 900cm³. What is the pressure at this position?

.....
[2]

Work & Power

1. A cyclist moves along a flat road against a resistive force of 100N. If the cyclist travels 100m calculate the work done by the cyclist
.....
.....[1]
2. Joan lifts a parcel of weight 100N onto a shelf that is 2m above the ground
 - a) Calculate the work done lifting the parcel onto the shelf
.....[1]
 - b) What type of energy does the parcel gain? [1]
3. Luke cycles a distance of 2000m against air resistance of 150N. He travels this distance in 7 minutes.
 - a) Calculate the work done by Luke
.....
.....[1]
 - b) What is Luke's Power output?
.....
.....[1]
4. The diagram shows a pumped storage system used to store water in a dam.

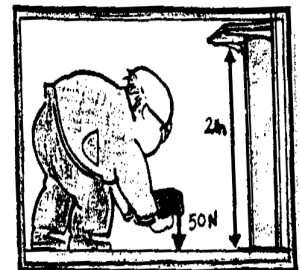


- a) Calculate the work done in pumping 10000N of water from the lower to the upper reservoir
.....
.....[1]
 - b) If it takes 10s to move 10000N of water from the top to the bottom, calculate the power output of the pump
.....
.....[1]
5. The power output of a crane is 1.7kW. Calculate how long it will take to lift a load of 5000N through a distance of 7600cm.
.....
.....[1]

6. This man lifts a brick from the floor up to a shelf. The brick has a mass of 5 kg and the shelf is 2m high.
 - a) What is the opposing force which tries to stop the man from lifting the brick?
.....[1]

- b) Calculate the weight of the brick.
.....[1]

- c) Calculate the Work done by the man.
.....[1]



Kinetic & Potential Energy

1. For the following pairs of objects state which has the most Kinetic Energy and by how much.

a) A car of mass 1000kg *or* a truck of mass 4300kg both moving at 10m/s

.....
[1]

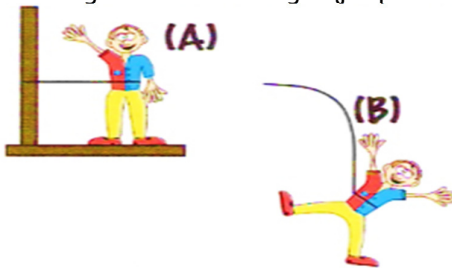
b) A car of mass 980kg moving at 7m/s *or* the same car moving at 16m/s

.....
[1]

2. The kinetic energy of a cyclist is 5000J. If the cyclist has a mass of 75kg and the bicycle has a mass of 18kg, calculate the speed the cyclist is going

.....
[2]

3. The diagram shows a bungee jumper at various stages:



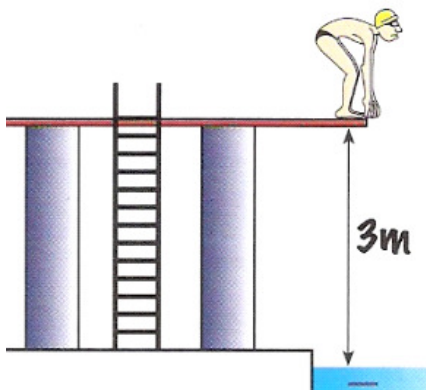
a) Which type of energy does he have at A?.....[1]

b) Which type of energy does he have while he is falling (B)?
[1]

c) If the bungee jumper has a mass of 55kg and is 5 up, calculate his potential energy

.....[1]

4. The diagram shows a diver. His mass is 72kg.



a) Calculate the weight of the diver
[1]

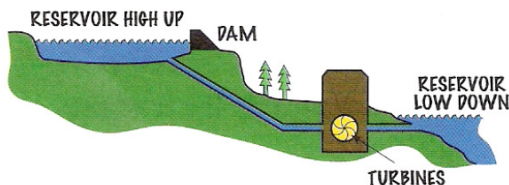
b) Calculate the gain in potential energy at the top of the board

[1]

c) Calculate the speed of the diver just before he touches the water

[2]

5. The diagram shows a hydroelectric power plant. The upper reservoir is 200 above the turbines and has a contains 10 million tonnes of water.



a) What is the total potential energy in the upper reservoir?

.....
[1]

b) 10000kg passes through the pipes per second. What would the speed of this water be when it passes through the turbines?

.....

[2]

Basic Principles of Waves

1. Name the two different types of waves and explain the differences

.....
[1]

2. Define the following and give the S.I. unit of each:

a) Amplitude

.....[1]

b) Wavelength

.....[1]

c) Frequency

.....[1]

d) Periodic time

.....[1]

3. The diagram shows a water wave. 8 waves are produced every 4 seconds

i) Calculate the wavelength and show with a λ on the diagram

.....[2]

ii) Calculate the frequency

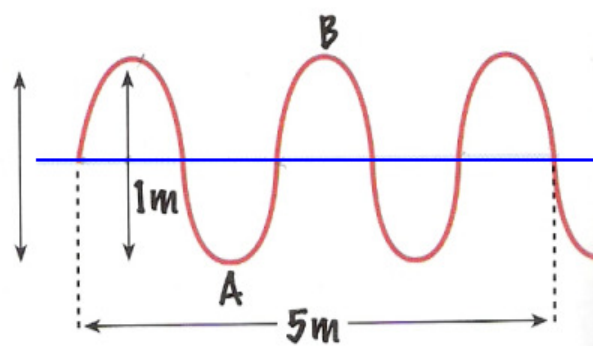
.....[2]

iii) What is the amplitude?

.....[2]

iv) Name the two points A and B

A..... B.....[1]



4. The diagram shows a wave

a) What type of wave is this?

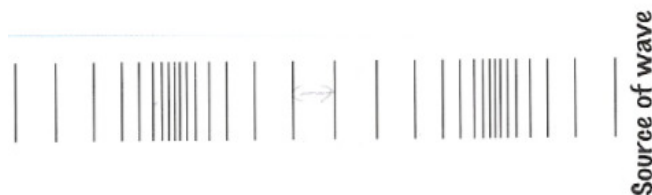
.....[1]

b) Draw the direction the wave is moving on the diagram, [1]

c) Mark the wavelength on the diagram [1]

d) If the source of the wave is vibrating to and fro 120 times every minute, what is the frequency of the wave?

.....[1]



5. The picture shows Lina making waves with a rope:



a) What type of wave is produced?

.....[1]

b) The frequency of the wave is 2Hz. Explain.

.....[1]

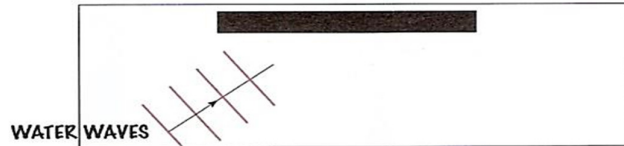
c) Show the amplitude and the wavelength on the diagram

d) The wavelength is 2m. Calculate the speed of the wave

.....[1]

Reflection, Refraction & Diffraction

1. The diagram shows a ripple tank. There is a barrier. Complete the diagram to show reflection of waves. [1]



2. The diagram shows water waves going from deep water to shallow water

a) Continue the path as they go into shallow water and back into the deep water [2]

b) What is this effect called?

.....[1]

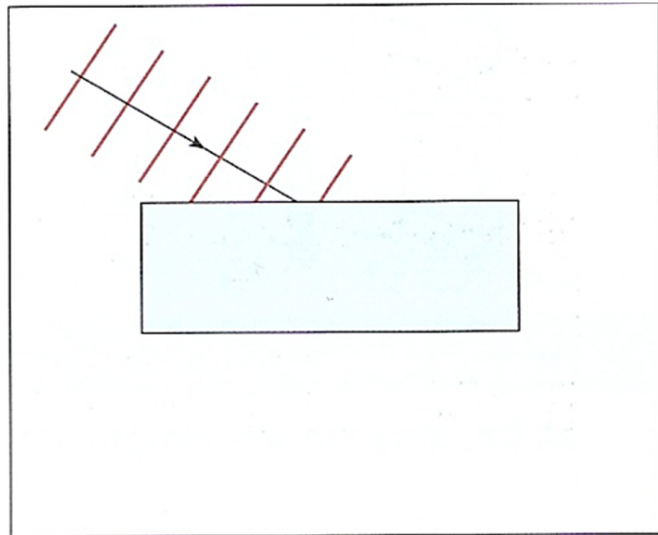
c) Explain why a wave behaves this way. (In your answer state what happens to frequency, speed, wavelength and amplitude of the wave)

.....

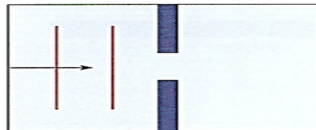
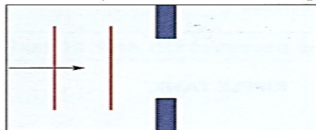
.....

.....

.....[2]



3. Complete the following diagrams [6]



4. ALL Electromagnetic waves travel through air at 300 million m/s

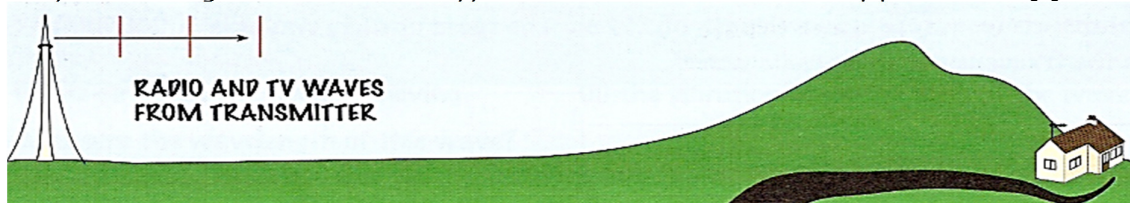
a) Calculate the wavelength of a radio station that broadcasts at a frequency of 900kHz

.....[1]

b) Calculate the frequency of a TV station which has waves of wavelength 60cm.

.....[1]

c) Complete the diagram to show what happens to the radio waves when they pass the hill [1]



d) What is the name of the effect that occurs on (c)?[1]

e) Explain why the people in the house would be unlikely to receive TV signals from (b)

.....[1]

Electromagnetic Spectrum

1. The electromagnetic spectrum is a group of waves

--	--	--	--	--	--	--

a) Label A to F with the correct type of electromagnetic wave, starting with the shortest wavelength. Also write down their wavelength and frequency [3]

b) Name two common properties of all electromagnetic waves.

.....[1]

2. This table is all mixed up! Match the correct letter (effects on living tissue) and number(uses) with each wave

	Effects on living tissue		Type of wave			Uses
A.	Probably none		<i>Gamma</i>		1.	- Communication - Broadcasting - Radar
B.	Heating of water in tissues can cause burning		<i>X - Ray</i>		2.	Imaging of internal structure in the body
C.	Causes burning of tissue		<i>Ultra Violet</i>		3.	- Fluorescent tubes - Tanning - Checking money
D.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous - Causes tanning		<i>Visible Light</i>		4.	- Kill bacteria in food - Sterilise medical equipment - Treat tumours
E.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous		<i>Infra Red</i>		5.	- Satellite communication - Cooking
F.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous - Kills cancer cells		<i>Microwave</i>		6.	- Radiant heaters - Grills - Remote controls - Night vision
G.	Activates sensitive cells in the retina		<i>Radio</i>		7.	- Seeing - Optical fibres

3. X-rays are part of the EM spectrum.

a) Explain the properties of x-rays which make them useful for us

.....[1]

b) Explain why x-rays can be dangerous to human beings

.....[1]

c) Microwaves are also part of the EM spectrum

i) Give one difference between x-rays and microwaves

.....[1]

ii) Name two uses for microwaves

.....[1]

iii) Recently there has been a lot of concern about the use of mobile phones and its effect on humans. Explain why microwaves could be harmful

.....[1]

Pitch & loudness

1. The picture shows 'Cikka the Rocker' on her world tour



a) What type of wave is a sound wave?[1]

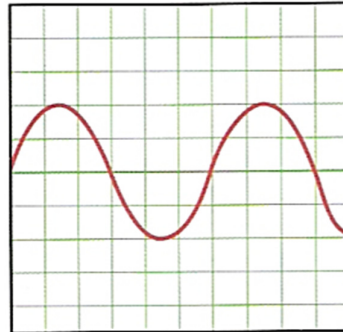
b) Describe how sound travels from her loudspeaker to the audience

.....

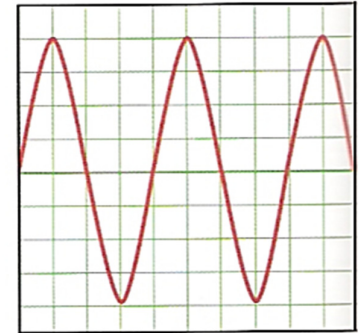
[2]

- c) During the sound check the sound engineer recorded two notes from her guitar. The traces from the CRO are shown in A and B.

Trace A



Trace B



- i) Which note is the **loudest**? Explain.

.....
[2]

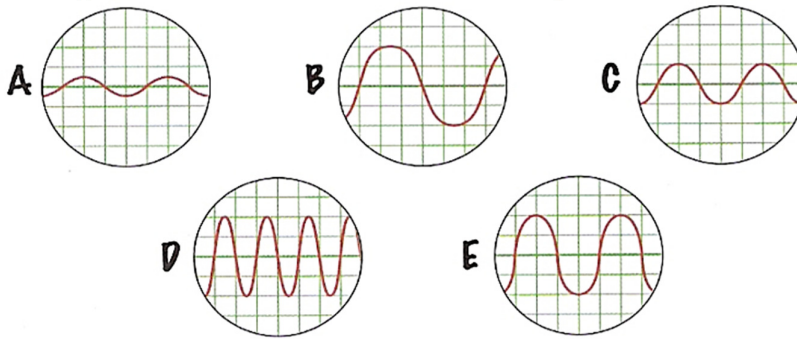
- ii) Which has the highest pitch? Explain

.....
[2]

- d) Cikka then played a note whose loudness AND pitch were doubled compared to the note in trace A. Sketch the trace of this note on A [2]

- e) Cikka then played a note whose loudness is a quarter and pitch a half of the note in trace B. Sketch the trace of this note on B. [2]

2. The diagram below shows traces on an oscilloscope screen:



Which show:

- a) The quietest sound
 b) The highest pitch
 c) The loudest sound
 d) The lowest pitch

[4]

3. Choose the correct words:

Increasing the (**frequency / amplitude**) of a sound will (**raise / lower**) the pitch of the sound, producing a (**higher / lower**) note. Decreasing the (**frequency / amplitude**) will (**raise / lower**) the pitch of the sound. Increasing the amplitude will increase the (**pitch / loudness**) of the sound and decreasing the amplitude will make the sound (**louder / quieter / higher / lower**).

[7]

Ultrasound

1. Complete the following:
 Sounds above 20,000Hz have too high a to be heard by the human ear. Sounds above this frequency are called [2]

2. Pregnant women have ultrasound scans to check on the progress of an unborn baby

a) What is ultrasound?

.....[1]

b) The image produced by ultrasound is not as clear as x-rays, so why is ultrasound used for looking at unborn babies rather than x-rays?

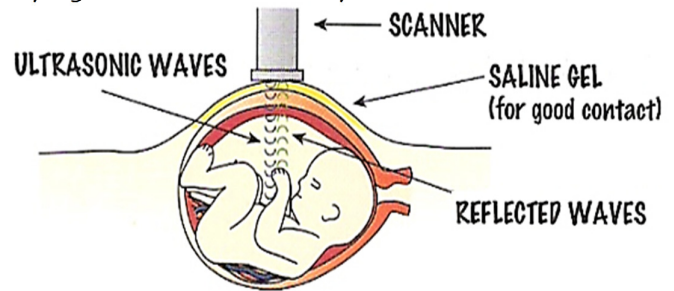
.....

.....[1]

c) Describe how ultrasound is used to produce an image of the baby

.....

.....[1]



3. Eco-sounders can be used to locate submarines

Using the diagram explain how eco-sounders are used to locate sub marines

.....

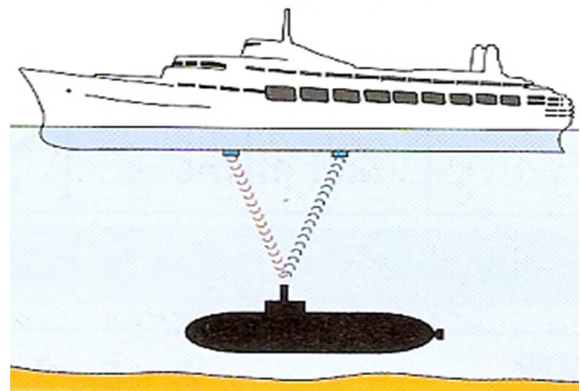
.....

.....

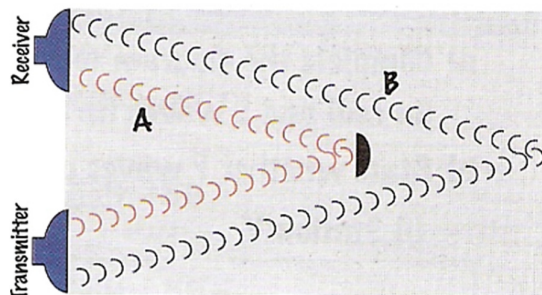
.....

.....

.....[3]



4. Ultrasound can be used to detect flaws in a material. An arrangement for this is shown below:



a) Which wave would arrive first to the receiver?[1]

b) Explain how this arrangement works to detect flaws.

.....

.....

.....

.....[1]

5. Explain how ultrasound can be used by dentists to clean plaque from teeth.

.....

.....[1]

Sound Calculations

1. A spectator on a sports field is 200m from the start. She see the starting gun fire and then hears the sound of the shot 0.6s later.
 - a) Calculate the speed of the sound
.....[1]
 - b) What will be the wavelength if the frequency if the sound is 200Hz?
.....[1]
 - c) Another sound wave has a frequency of 2kHz. How does the frequency compare to the wave in part (b)? What will the ratios of their *wavelengths* be?
.....
.....[1]
2. Sound travels through some things but not others
 - a) Which of the following can sound travel through? [4]

	Solids
--	--------

	Vacuum
--	--------

	Liquids
--	---------

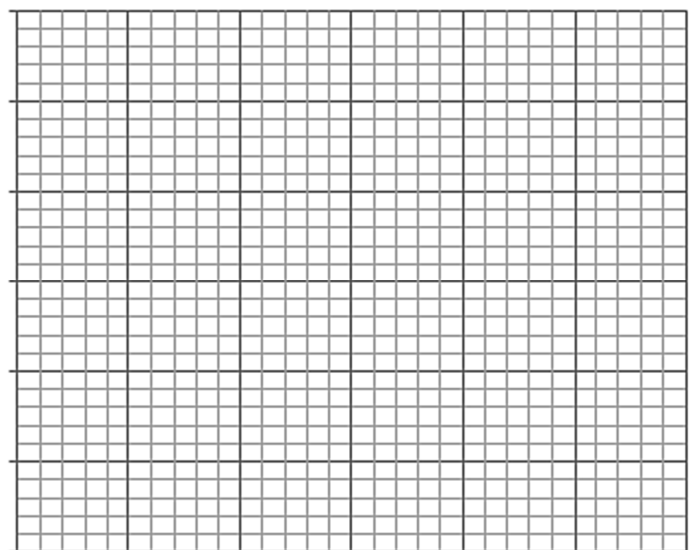
	Gases
--	-------
 - b) Explain your answers
.....[1]
 - c) Which medium in a) does sound travel fastest in? Explain why
.....[2]
3. Bats use ultrasound to catch their prey. A typical victim would be a mosquito or a moth. A bat cannot sense anything smaller than the wavelength of sound it uses. What frequency does the bat need to send out to help catch a moth? (Hint: estimate the size of a moth first)
.....
.....[3]



4. Some students are doing an experiment to measure the speed of sound. They are positioned at 200m intervals across the field. Each has a stopwatch. The student with the starting pistol simultaneously pulls the trigger and drops his arm. The students START their stopwatches when they see the arm fall, and STOP the watches when they hear the bang. Their times are shown below:

Distance (m)	200	400	600	800	1000
Time (s)	0.9	1.2	1.8	2.4	3.0

- a) Plot a graph of distance against time [3]
- b) Use your graph to calculate the speed of time
.....
.....[2]
- c) One point does not quite fit the pattern. Why might you expect this point to be the one least accurately measured?
.....
.....[1]

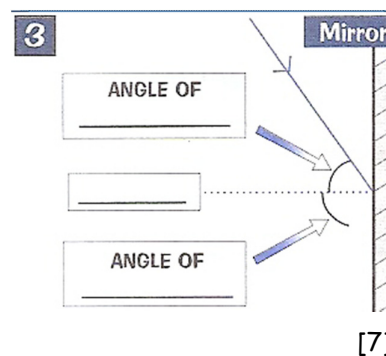
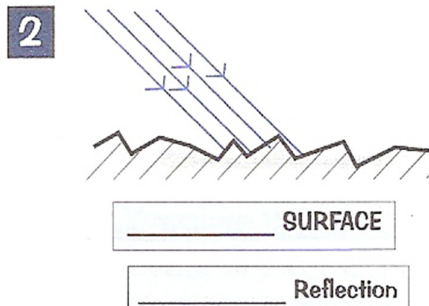
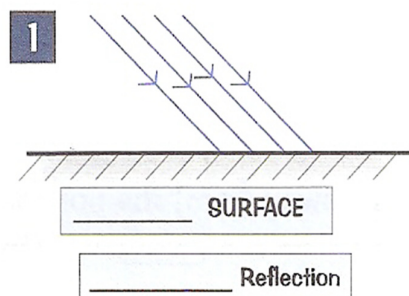


Reflection

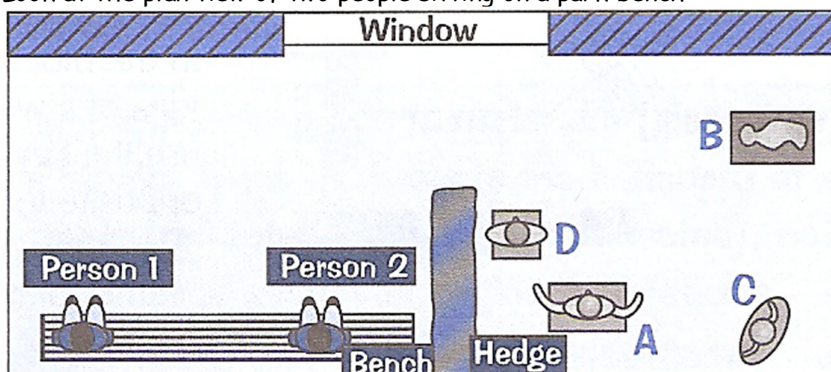
1. Fill in the blanks:

- Some objects give out their own light. These are called objects. We see all other objects because they light. These are called objects. [3]
- Some objects reflect light without sending it off in many different directions. This is called a reflection and objects which do this produce a [3]
- Most objects send the reflected light in many different directions giving a reflection. These objects do not produce a image [2]
- The law of reflection states that "the angle of is to the angle of [3]

2. The diagrams show rays arriving at a surface. Complete the labels and draw the reflected rays :



3. Look at the plan view of two people sitting on a park bench



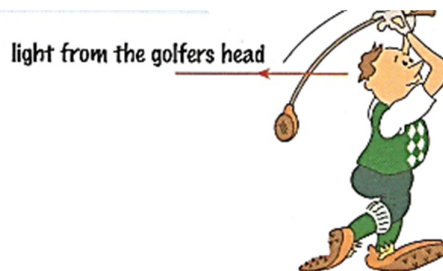
They can see some statues reflected in the window.

Use the law of reflection to decide which of the statues each can see

Person 1:[2]

Person 2:[2]

4. The diagram shows a periscope with two mirrors. Complete the path of the ray through the periscope.



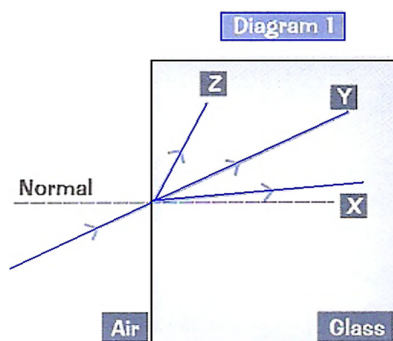
[1]

Refraction

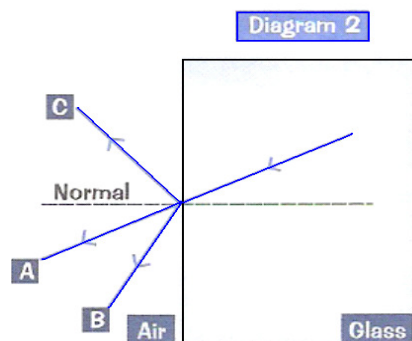
1. Fill in the blanks:

- Light travels at different in different media (e.g. air to glass)
- The speed of light will when it travels to a denser medium
- When the light goes back into air its speed will
- The normal is an imaginary line which is at to the surface
- The frequency of light when light travels through a denser medium [5]

2. In the diagram 1, a ray *enters* a glass block. Which ray; X, Y or Z shows how it will continue? [1]

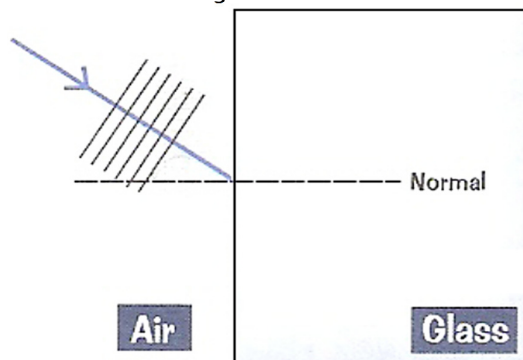


3. In the diagram 2 a ray *leaves* the block. Which ray; A, B or C shows the correct path?



- When a ray of light enters a glass block it (*slows down / speeds up*) and is bent (*towards / away from*) the normal.
- When a ray of light leaves a glass block it (*slows down / speeds up*) and is bent (*towards / away from*) the normal.

6. Look at the diagram



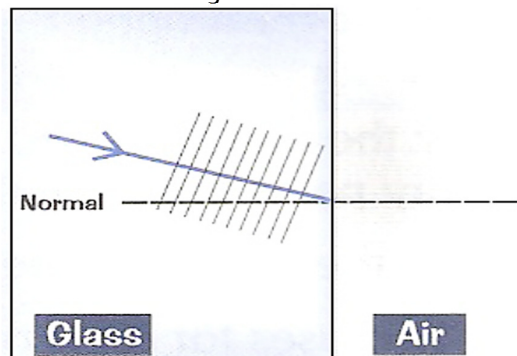
Write an explanation of the refraction of this ray in the block using the words:

wavefront, angle to the normal, slows down, wavelength, direction, frequency, unchanged

.....

[3]

7. Look at the diagram



Write an explanation of the refraction of this ray as it exits the block using the words:

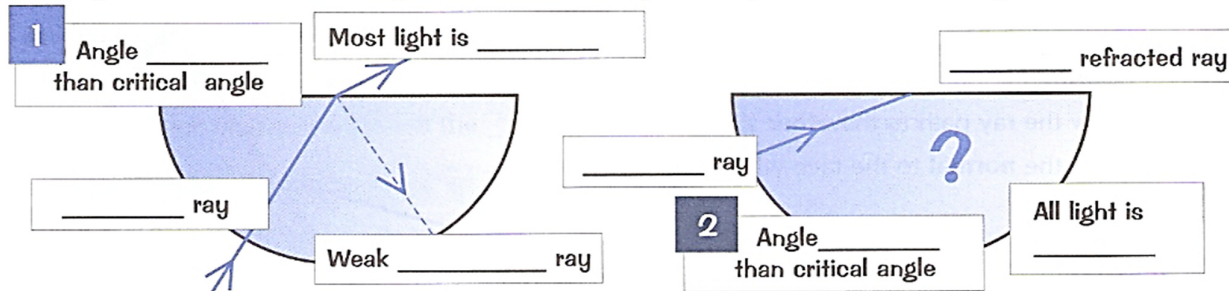
wavefront, angle to the normal, speeds up, wavelength, direction, frequency, unchanged

.....

[3]

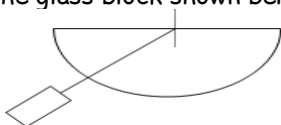
Total Internal Reflection

1. The diagram shows two identical glass blocks with a ray entering at two different angles



- a) Complete the labelling for diagram 1 and diagram 2. [8]
b) Complete diagram 2 [2]

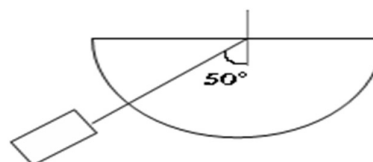
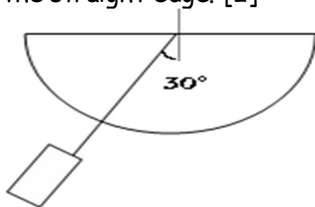
2. Joseph sets up the glass block shown below.



- a) Look at the diagram and label :
i) the normal (N)
ii) the angle of incidence (i)

- b) Explain why the ray of light passes straight through and is not deviated when entering the glass block[1]

- c) The critical angle of glass is 42° . The diagrams below show light incident at an angle of 30° in the first figure and 50° in the second figure. Complete the diagrams, showing the path the ray follows after it hits the straight edge. [2]



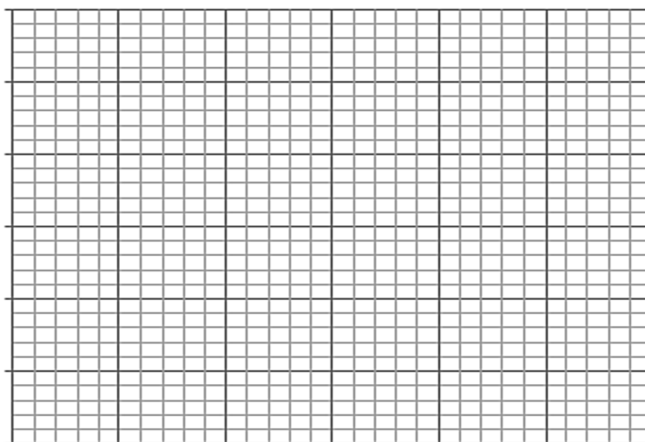
- d) Complete:

- i) In the first figure the ray of light is[1]
ii) In the second figure the ray of light is[1]

3. The following table of results is obtained from an experiment to find the critical angle of a semicircular plastic block.

Angle in plastic	0	10	20	30	40	45	50
Angle in air	0	13	27	41	57	68	90

- a) Plot a graph of angle in air (y-axis) against angle in plastic (x-axis). Draw the best smooth curve. [3]
b) From your graph find the angle in air when the angle in plastic is 35° [1]
c) From your graph find the angle in plastic when the angle in air is 25° [1]
d) What is the critical angle of this kind of plastic? [1]



Uses of Total Internal Reflection

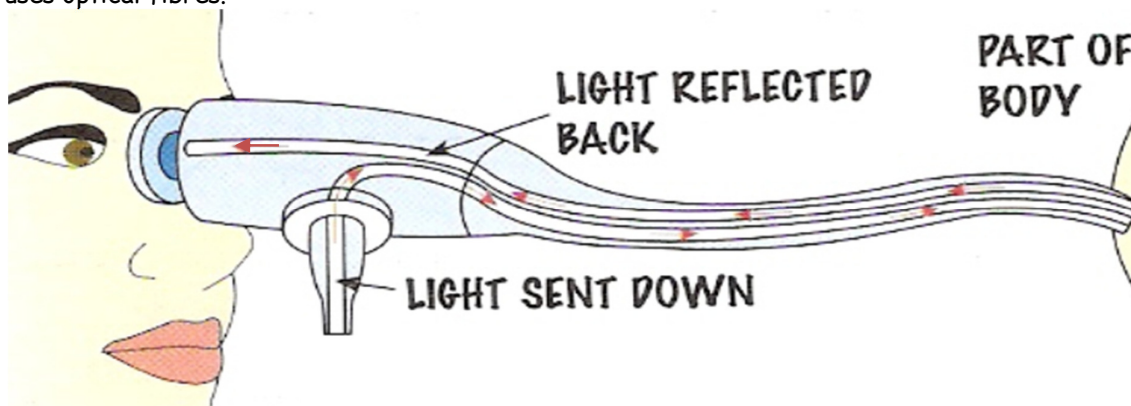
1. The diagram shows a length of optical fibre. Its critical angle is 42° .



- a) Complete the passage of the ray through the optical fibre. Make sure to mark the angles of incidence and reflection [2]
 b) Fibre optics can be used for telecommunications. Explain the advantages over electrical signals

.....[2]

2. An endoscope is an instrument that can be used by doctors to see inside a patients knee. This instrument uses optical fibres.



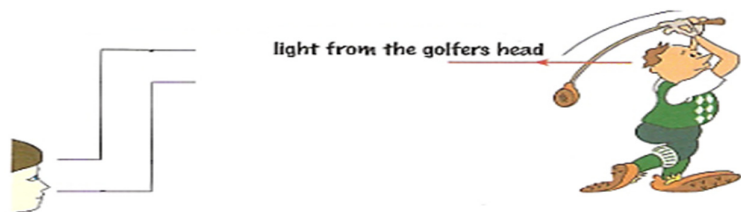
Explain how the endoscope works:

.....[2]

3. Reflecting prism are used in periscopes

- a) Complete the diagram to show:

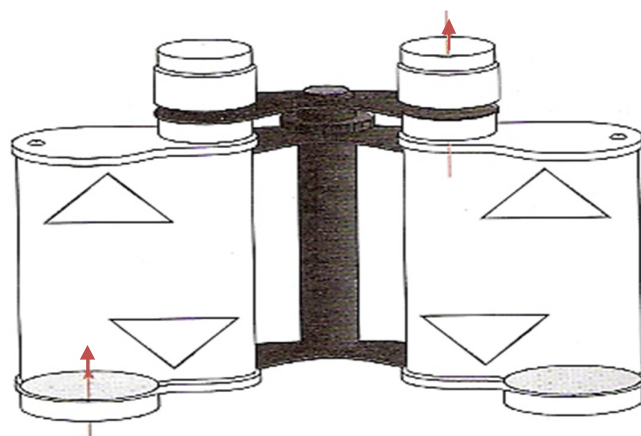
- i) The arrangement of the prisms [1]
 ii) The path of the ray of light



- b) They are also used in prism binoculars. Complete the path of light [2]

- c) Explain why the light is reflected in the prism and not refracted, even though it is 'see through'

.....[2]



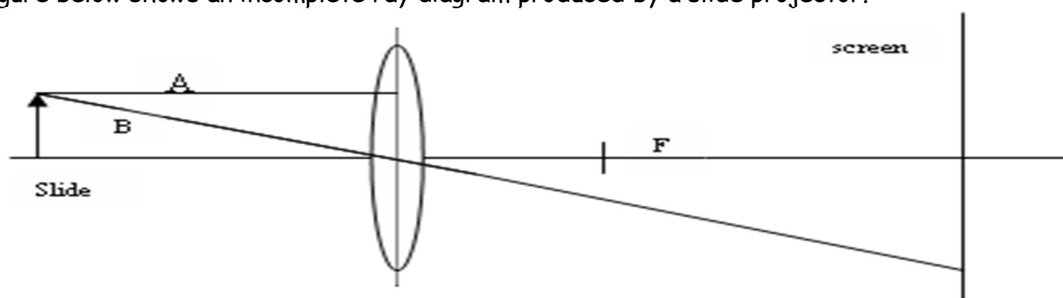
Lenses

1. An object of height 2 cm is placed 7 cm in front of a converging lens of focal length 3 cm. Draw a ray diagram to show this. Use an appropriate scale if needed. [3]

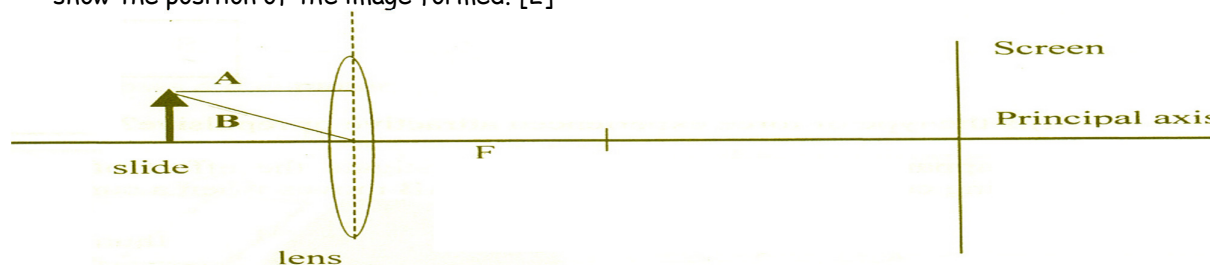
- What is the height of the image?[1]
- Name three properties of this image.[3]
- The object is moved 2 cm from the lens. Draw the new ray diagram [3]

- Name three properties of this image.[3]
- Suggest one practical use of an image formed in this way by a converging lens.[1]

2. The figure below shows an incomplete ray diagram produced by a slide projector.

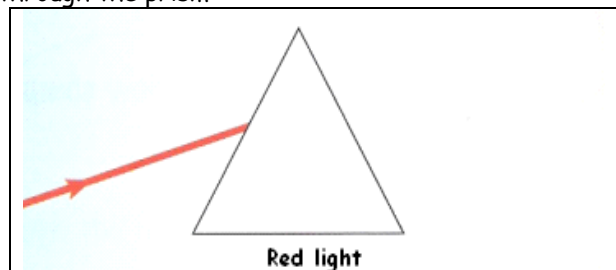
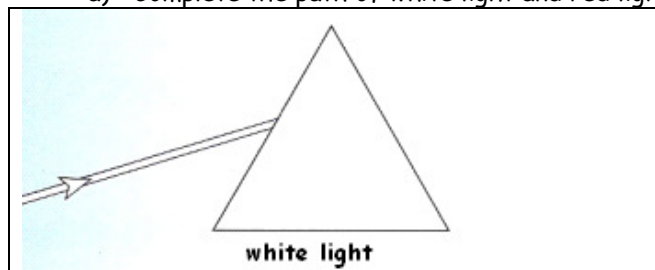


- Is the lens converging or diverging?[1]
- Draw the arrows on rays A and B, to show the direction followed by the light. [1]
- Complete ray A to show the image obtained on the screen. [1]
- Give three properties of the image.....[3]
- When the object was placed 8 cm in front of the lens, a focused image was obtained on a screen placed 2 m behind the lens. Work out the magnification.
.....[2]
- A detail on the slide is 12 mm high. Work out the size of the image formed.
.....[2]
- The slide is now moved towards the lens as shown in the diagram below. Complete the rays A and B, and show the position of the image formed. [2]



Dispersion of Light

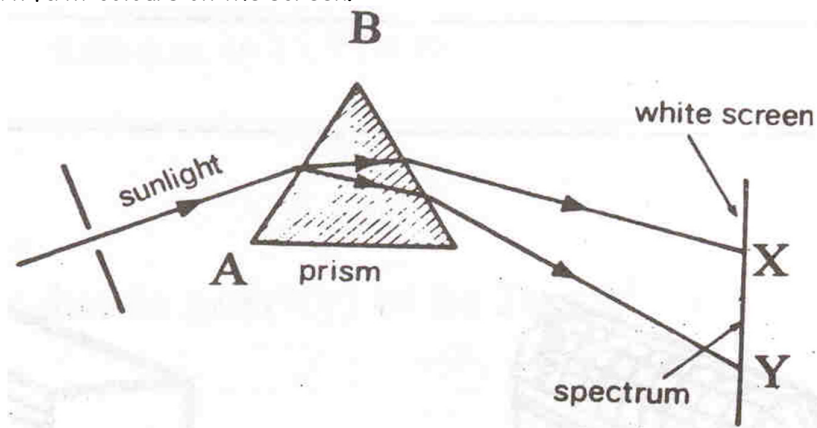
1. The drawing shows a triangular prism
a) Complete the path of white light and red light through the prism



- b) Explain why white light is split up by the prism

.....
.....[2]

2. By placing a prism as shown in the diagram and shining a narrow beam of white light on the face AB, Sarah finds she can obtain faint colours on the screen.



- a) Dispersion of white light results on the screen between X and Y. Explain the meaning of the word 'dispersion'.

.....
.....[2]

- b) Which colours are found at X and Y on the screen?

.....[2]

- c) A type of invisible radiation can be found beyond the end Y of the spectrum. What is it called?

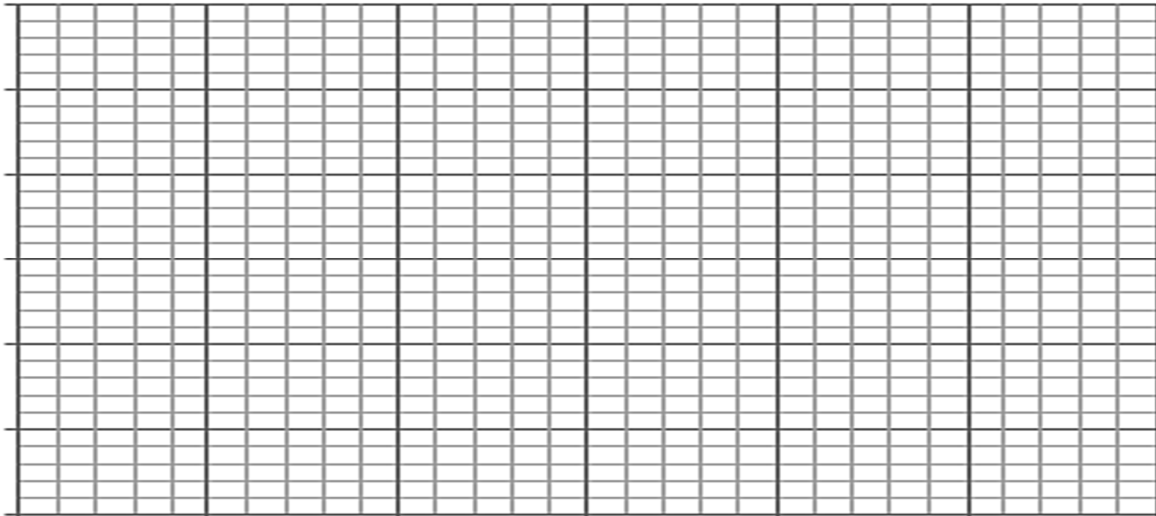
.....[2]

Speed and Velocity

- A car travels at 110km in 2hrs. Calculate its average speed in:
 - Km/h[1]
 - m/s[1]
- An aeroplane flies 4200km at an average speed of 1000km/h. Calculate the total distance travelled[1]
- A 1500m runner takes 3mins 16s to complete the race. Calculate his average speed.[1]
- A cyclist cycles at an average speed of 10m/s. Calculate the distance travelled in 3 hours[1]
- Kurt and Mirco were having an 800m race. Their times were recorded every 100m and the table shows the results.

		Distance run (m)								
		0	100	200	300	400	500	600	700	800
Time (s)	Mirco	0	15	30	45	65	85	105	130	155
	Kurt	0	20	40	60	80	95	110	125	140

- a) On the same axis draw both distance - time graphs [4]



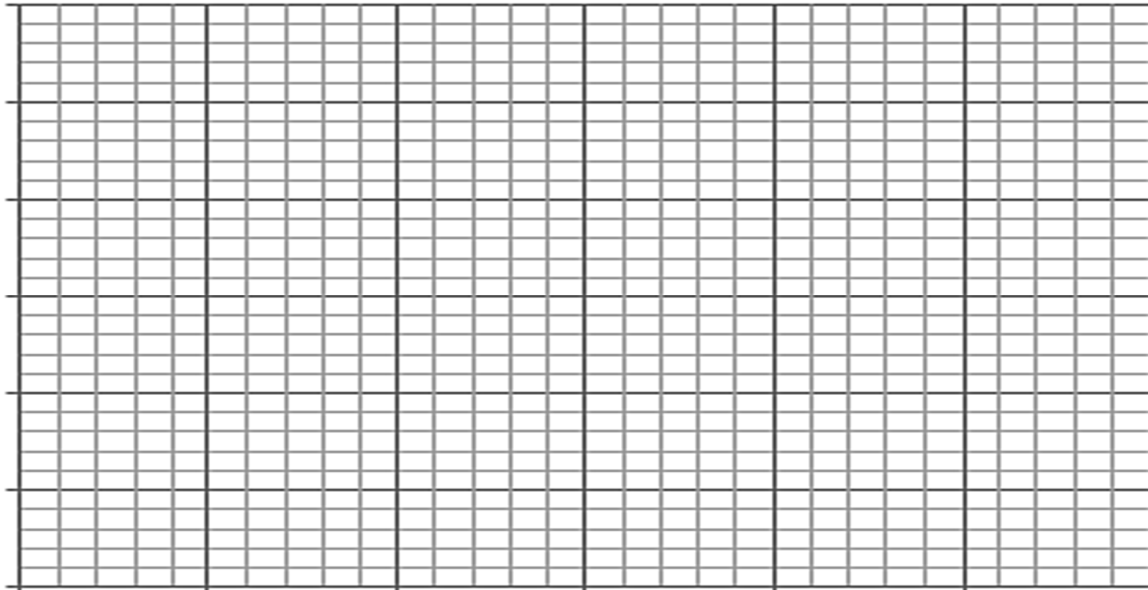
- b) Between which two times was Mirco's speed the slowest?[1]
- c) Between which two times was Kurt's speed the slowest?[1]
- d) Who won the race?[1]
- e) By how many seconds did he win the race?[1]

Acceleration

1. A car accelerates uniformly from rest to a speed of 12m/s in a time of 6s. Calculate the acceleration of the car.
.....[1]
2. A plane lands at a velocity of 35m/s. It is stopped in 2s. Calculate the deceleration of the plane
.....[1]
3. A train accelerates uniformly from rest at a rate of 1.7m/s^2 . If the train accelerates for 14s, calculate the final velocity of the train
.....[1]
4. A car accelerates from 15m/s to 25m/s in 7s. Calculate its acceleration
.....[1]
5. A cyclist at the top of a hill has a speed of 4m/s. He then accelerates at 2.3m/s^2 for 4s. Calculate his speed after 4s
.....[1]
6. The table below shows the velocity of a car which travels between two traffic lights:

Velocity (m/s)	0	5	10	10	10	13	16	16	12	8	4	0
Time (s)	0	2	4	6	8	10	12	14	16	18	20	22

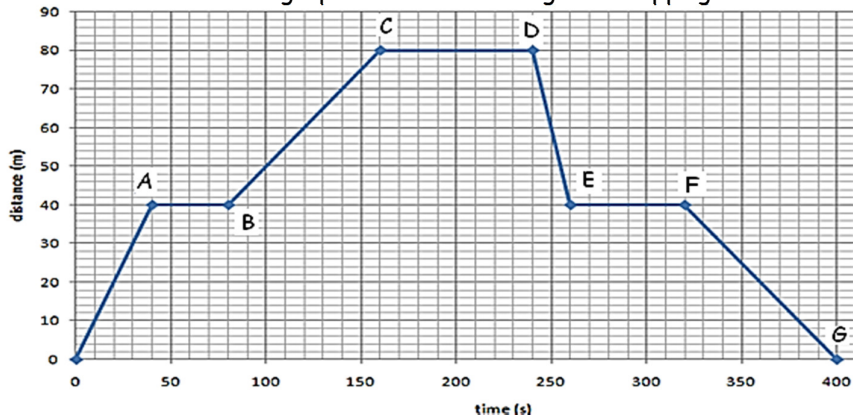
- a) Plot the velocity time graph:



- b) Between which two times was the car's acceleration the greatest?
.....[1]
- c) For how long did the car travel at constant velocity?
.....[1]

Motion Graphs

1. The distance-time graph shows Guza doing her shopping



- a) Describe the first four minutes of Guza's trip

.....

[1]

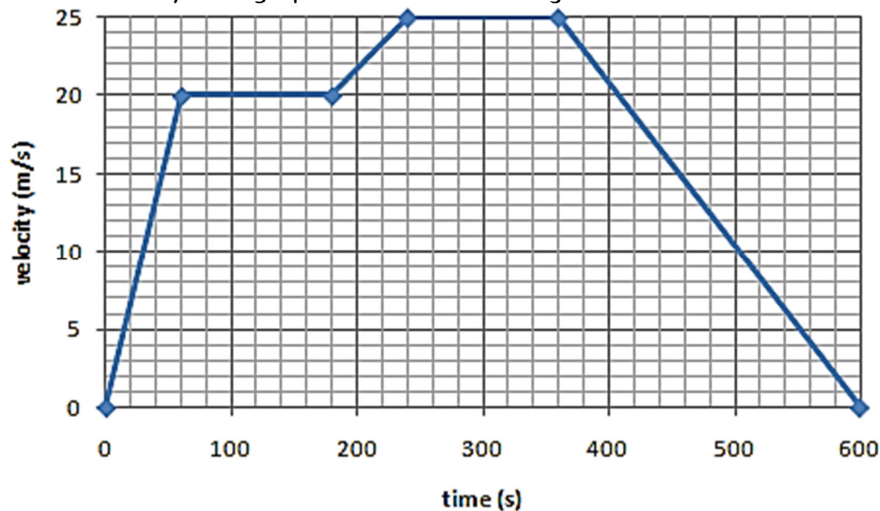
- b) In which region was Guza travelling the fastest?

.....[1]

- c) Calculate Guza's speed at OA, BC and DE

.....
[1]

2. The velocity-time graph is for a train moving from one station to the next.



- a) Describe the motion of the train

.....

[2]

- b) Calculate the 2 accelerations

.....

[2]

- c) Calculate the deceleration of the train

.....
[1]

- d) Calculate the total distance travelled.

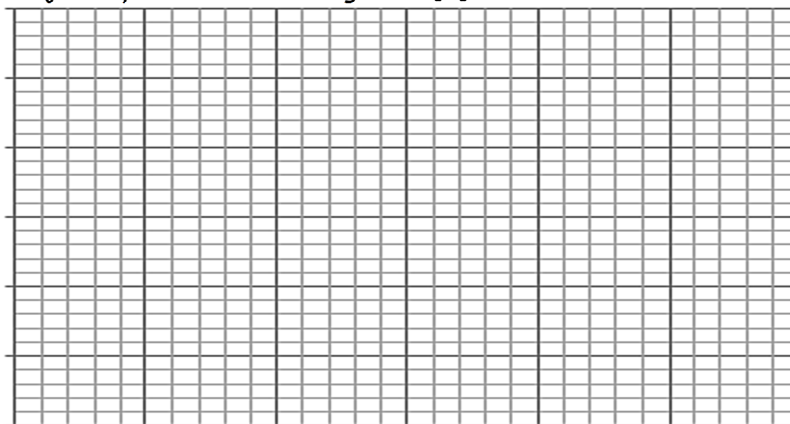
.....
[1]

3. Complete the table by stating the significance of features sometimes observed on distance-time and velocity-time graphs:

Feature on graph	Distance - Time	Velocity - Time
Gradient		
Horizontal Line		
Upward Line		
Downward Line		
Area under the Graph	NOT APPLICABLE	

4. Draw the velocity-time graph for a car journey from the following table [3]

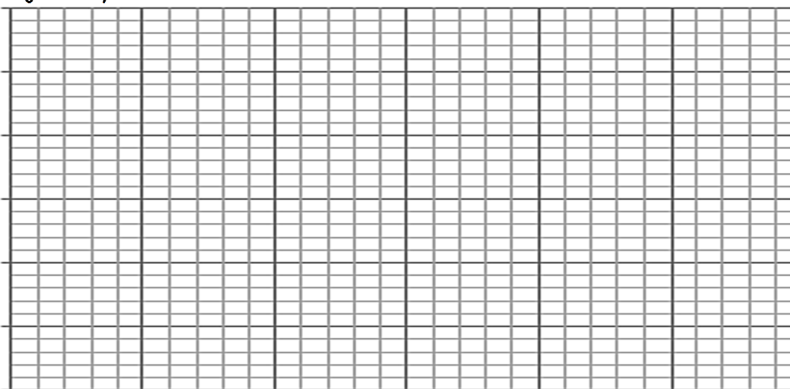
<i>Velocity (m/s)</i>	<i>Time (s)</i>
0	0
4	2
8	4
12	6
16	8
20	10
24	12
24	16
12	18
0	20



- a) Describe the motion of the car. (on graph) [2]
- b) Calculate the acceleration of the car in the first 12 seconds
[1]
- c) If the car had a mass of 1000kg what force was needed to produce the acceleration in (b)?
[1]
- d) Calculate the deceleration of the car in the last 4s.
[1]
- e) Calculate the total distance travelled by the car
[1]

5. Draw a distance-time graph of a bike journey:

<i>Distance (m)</i>	<i>Time (s)</i>
0	0
20	5
40	10
60	15
80	20
100	25
100	30
50	35
0	40



- a) Describe the motion of the bike for the whole journey (on graph) [2]
- b) Calculate the speed of the bike between the times of 20 and 25 seconds
[1]
- c) For how long is the bike stationary?[1]
- d) What is the total distance covered by the cyclist?
[1]

Stopping

1. What is meant by '*braking distance*'?.....[1]
2. What is meant by '*thinking distance*'?.....[1]
3. A car of mass 900 kg is traveling at 30 m/s. A small child runs in front. The driver's reaction time is 0.6s and it takes a further 2s to brake the car to a halt. Calculate:
 - (a) the thinking distance traveled by the car during the reaction time.
.....[1]
 - (b) the braking deceleration.
.....[1]
 - (c) the average braking force.....[1]
 - (d) the change in K.E.....[1]
 - (e) the work done by the brakes.
.....[1]
 - (f) Sketch a speed - time graph for the car.;
 - (g) Explain how the graph will change if:
 - i) the driver drinks alcohol[1]
 - ii) the road is wet.[1]
4. A driver of a car has a thinking time of 0.7s, that is, there is a delay of 0.7s between the driver deciding to stop the car and pressing the brake pedal.
 - (a) If the car is travelling at 20m/s calculate the distance covered by the car during the thinking time.
.....[1]
 - (b) Once the brake pedal has been pressed, the car decelerates uniformly and stops in 3s. Draw a speed-time graph to show how the speed of the car changes during the last 3s.
 - (c) From your graph find the distance covered by the car during braking.
.....[1]
 - (d) Calculate the total distance covered by the car, between the driver deciding to stop the car and the car finally coming to rest.
.....[1]

Equations of Motion

1. What quantities do the following symbols stand for in the equations of motion:

$s =$ [1]

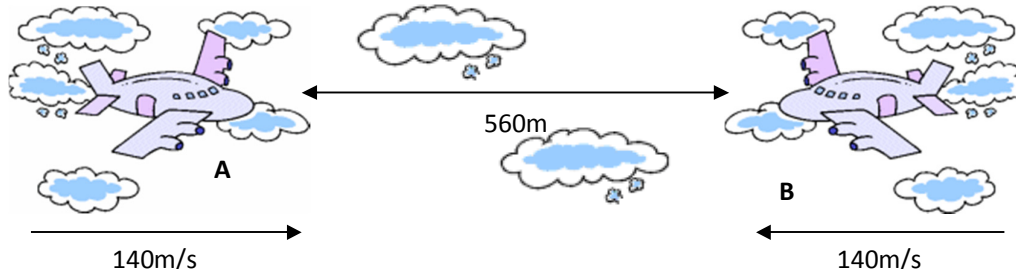
$u =$ [1]

$v =$ [1]

$t =$ [1]

$a =$ [1]

2. Two planes are shown below:



- a) The planes do not have the same velocity. Explain why.

.....[1]

- b) If the planes continue at their current velocities, how long would it be before they collide?

..... [2]

- c) To avoid collision, plane A climbs away and plane B decelerates at 20m/s^2 . What will the velocity of plane B be after 2s of deceleration?

.....[2]

- d) A third plane is on the runway. It needs to reach a takeoff speed of 120m/s . If it takes 7.5s to reach its takeoff speed from rest, what is its acceleration?

.....[2]

3. A sprinter accelerates uniformly from rest during the first 30m of a 100m race.

- a) What is the athlete's initial velocity?[1]

- b) The athlete takes 4.8s to run the first 30m.

- i) What is the final speed at the end of the 30m?

..... [2]

- ii) Calculate his acceleration.

..... [2]

- c) The athlete then runs the rest of the race at uniform velocity.

- i) Calculate the time he takes to run the rest of the race.

..... [2]

- ii) What is the total time to run the race?

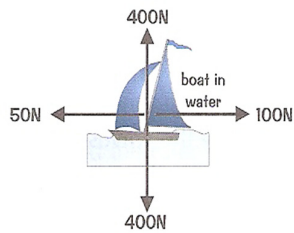
..... [2]

Equations of Motion – free fall

1. What is the meaning of free fall?
.....
.....[2]
2. What is the acceleration when an object is free-falling?[1]
3. What is the initial velocity when an object is free-falling?[1]
4. Mother drops the keys from the roof of her house, 9m above the ground.
 - a) What is the value of the initial velocity of the keys?[1]
 - b) Neglecting air resistance, state the value of the acceleration of the keys.[1]
 - c) Work out how long the keys take to reach the ground.
.....
..... [2]
 - d) Calculate the velocity with which the keys hit the ground.
.....
..... [2]
 - e) Kevin suggests to his mother that she ties a handkerchief to the keys so that it acts like a parachute. How would the handkerchief affect the time the keys take to reach the ground.
.....
.....[2]
5. A person drops a heavy object from a helicopter hovering at 300m. During the fall there is no air resistance.
 - a) What is the initial vertical velocity of the object?[1]
 - b) What is the vertical acceleration of the object?[1]
 - c) What is the vertical distance moved by the object before it hits the ground?
.....
.....[2]
 - d) Calculate the time taken by the object to reach the ground.
.....
.....[2]
 - e) Calculate the vertical velocity with which the object hits the ground.
.....
.....[2]

Terminal Velocity

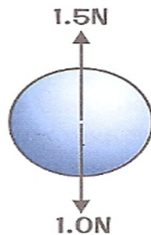
1. Calculate the resultant force of each diagram. Also state if the body is accelerating or not and in which direction:



A boat in the water

.....

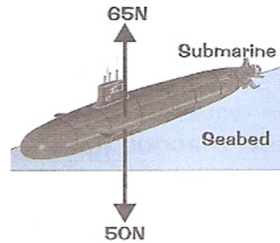
.....[2]



A ball being blown upwards by a jet of air

.....

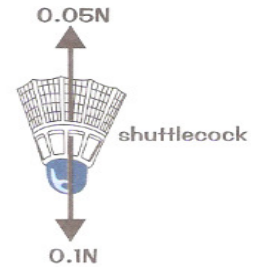
.....[2]



A model submarine on the sea bed

.....

.....[2]



A shuttlecock in the air

.....

.....[2]

2. The drawing shows a skydiver of mass 68kg.



i) What is the weight of the sky diver?[1]

ii) What happens to the weight as the diver falls?[1]

iii) As the sky diver steps out of the plane she will fall and accelerate. Explain in terms of the forces acting on the diver, why she accelerates

.....

.....[2]

iv) What will happen to the air resistance as she falls?[1]

v) Explain why the skydiver will eventually stop accelerating.

.....

.....[2]

vi) What is the name given to this velocity?[1]

vii) Eventually the skydiver will open her parachute

i) When should she open the parachute? Why?

.....

.....[2]

ii) What effect will this have on the air resistance?[1]

iii) What will happen to the velocity of the parachutist now?[1]

3. Explain, using diagrams what terminal velocity is.

.....

.....

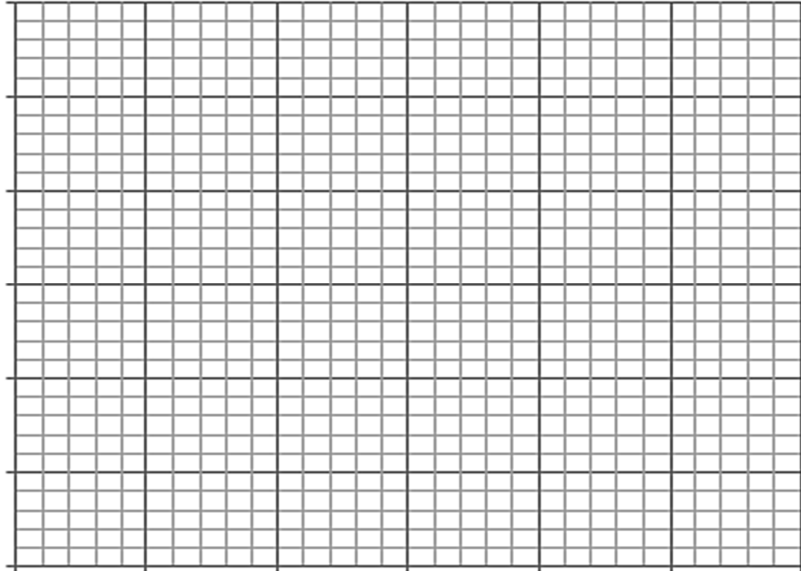
.....

.....

.....[4]

4. Plot a graph of velocity (m/s) on the y-axis against time (s) showing the motion of a skydiver after stepping out of an aeroplane.

Velocity (m/s)	Time (s)
0	0
4.5	2
16.5	4
23.0	6
29.0	8
36.0	10
43.5	12
50.0	14
56.0	16
60.0	18
60.0	20



- a) Find the terminal velocity of the skydiver[1]
- b) Estimate the velocity of the skydiver after
- i) 5s[1]
- ii) 12.5s.....[1]
- c) At what time does the skydiver reach terminal velocity?[1]
- d) The skydiver now opens the parachute 20s after jumping out of the aeroplane. Describe the extra force acting on her and its effect upon her speed
-
-[2]
- e) Will the skydiver reach a new terminal velocity? Explain
-
-[2]

5. Choose from the words supplied to fill in the blanks in the paragraph below:

<i>Decelerates</i>	<i>Decrease</i>	<i>Less</i>	<i>Balances</i>
<i>Increase</i>	<i>Constant</i>	<i>Greater</i>	<i>accelerates</i>

When a skydiver jumps out of a plane, his weight is than his air resistance, so he downwards. This causes his air resistance to until it his weight. At this point his velocity is When his parachute opens, his air resistance is than his weight, so he This causes his air resistance to until it his weight. Then his velocity is once again [10]

Collisions and Explosions

1. Place the following four trucks in order of increasing momentum:



Truck A
speed = 30 m/s
mass = 3000 kg



Truck B
speed = 10 m/s
mass = 4500 kg



Truck C
speed = 20 m/s
mass = 4000 kg



Truck D
speed = 15 m/s
mass = 3500 kg

.....
.....[2]

2. A truck of mass 1750kg moving at 50km/h collides with a car of mass 900kg moving in the same direction at 8m/s. If, after collision the car moves off with a velocity of 11m/s calculate the velocity of the truck after collision

.....
.....
.....
.....[2]

3. If the car in question 2 was moving in the opposite direction of the truck, calculate the *combined* velocity of both vehicles after collision

.....
.....
.....
.....[2]

4. A car of mass 1300kg travelling at 10m/s collides into a van of mass 1800kg which was at rest.

a) Calculate the total momentum before collision

.....[1]

b) What is the total momentum after collision?

.....[1]

c) If the car and van move off together after collision, calculate their combined velocity

.....
.....
.....
.....[2]

d) Calculate the total kinetic energy before collision

.....[1]

e) Calculate the total kinetic energy after collision

.....[1]

f) Why is the KE different before and after collision?

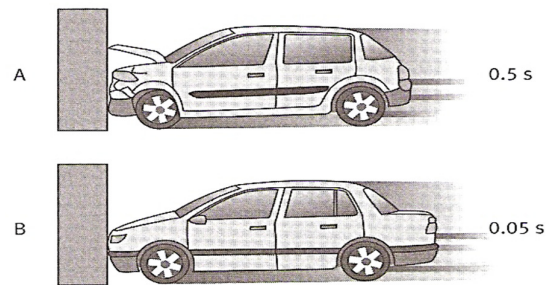
.....[1]

5. A skateboarder of mass 60kg is moving at 6m/s. He skates past his bag, picks it up from the floor and slows down to 4.3m/s. Calculate the mass of the skater's bag
-
-
-
-[2]
6. A canon ball of mass 2kg is fired at a velocity of 150m/s from a stationary canon of mass 600kg.
- a) What is the TOTAL momentum before the canon is fired?
-[1]
- b) What is the TOTAL momentum after the canon is fired?
-[1]
- c) Calculate the momentum of the canon ball just after firing
-[1]
- d) What is the momentum of the canon just after firing
-[1]
- e) Calculate the velocity at which the canon moves after firing (show direction)
-
-
-
-[2]
7. Pete fires a gun towards a sack full of sand. A bullet of mass 0.012kg is fired at a velocity of 710m/s.
- a) Calculate the momentum with which the bullet leaves the gun.
-[1]
- b) Pete is observed to move backwards. Explain why this happens.
-[1]
- c) What is the value of Pete's momentum?
-[1]
- d) Calculate the velocity with which Pete moves back if his mass is 90kg and the gun's mass is 6kg
-
-
-
-[2]
- e) The bullet hits a sack of sand of mass 4.5kg at close range with a velocity of 500m/s. Calculate the velocity with which the sack and bullet move on together after collision.
-
-
-
-[2]

Impulse and safety

- Sam and Jo are two skaters. Sam pushes Jo who was at rest. Sam exerts a force on Jo which increases her momentum.
 - Sam could increase Jo's momentum more if he:
 - Pushes her with *more/less* force
 - Pushes her for a *longer/shorter* time [2]
 - Write in the missing quantities and units:
 - Change of momentum (IMPULSE) = x
 b) = x (UNITS)
 - Jo gains 120kgm/s of momentum when Sam pushes her. Sam exerts a force on her while they are in contact. This lasts for 2 seconds. Calculate the size of the average force that Sam exerts on Jo
[1]
- Explain why:
 - You bend your legs when you jump down from a table rather than keep them rigid
[1]
 - You would prefer to fall on a bed than on concrete
[1]
- Modern cars are fitted with many safety features.
 - Why are car safety often designed to slow the car and passengers down over a *long time*?
[1]
 - How do the following features achieve this?
 - Crumple Zones.....[1]
 - Air bags.....[1]
- Look at the pictures of two crashed cars. Both cars were travelling at 20m/s when they hit the wall.
 - Draw lines to link each of the phrases with one of the pictures:

small time
big time
small force
big force



- Both of the driver's masses are 60kg. Calculate their momentum before collision.
[1]
- He is brought to rest by the force from the seatbelt. The times are shown on the pictures. Calculate the force on each driver.
[1]
- Which is the safer car. Explain

[1]

5. Imagine the driver in car B in question 4 had not been wearing a seatbelt.
- What object (or objects) would have provided the force to slow him down?
.....[1]
 - It is safer to have BOTH a seatbelt AND a crumple zone. Explain the disadvantages of
 - A seatbelt on its own[1]
 - A crumple zone on its own.....[1]
6. A 750kg car is travelling at 30m/s along the motorway. It crashes into the barrier of the central reservation and is stopped in a period of 1.2s.
- Find the size of the average force acting on the car stop it.
.....[1]
 - Explain in terms of the forces acting why the occupants of the car are likely to be less severely injured if they are wearing seat belts made of slightly *stretchy* material.
.....
.....[1]
7. A van of mass 1200kg was moving at a velocity of 8m/s when it was involved in a head on collision with a lorry moving in the opposite direction. Assuming that the van came to a stop after collision,
- Calculate the momentum of the van before collision
.....[1]
 - What is the momentum of the van after collision
.....[1]
 - If the van took 0.3s to stop, calculate the force that acted on the driver
.....[1]
 - Modern vehicles are designed to crumple gradually after impact. This means that the force acting on the driver would be spread out over a longer time. Calculate the force on the driver if the lorry stops in 0.8s.
.....[1]
8. High flying birds could be a source of danger to aircrafts because if they collide with the windscreen, the resulting impact could cause serious damage to the aircraft. 15 ducks, each of mass 1.3kg, travelling at 20m/s collide with an aeroplane of mass 2300kg. The plane is travelling in the opposite direction with a velocity of 212m/s
- Calculate the momentum of the plane before collision.
.....[1]
 - Calculate the momentum of the ducks before collision
.....[1]
 - Calculate the total momentum of the plane and ducks before collision
.....[1]
 - One of the ducks hits the plane's windscreen. Calculate the impact force on the windscreen, if the collision lasts for 0.0012s.
.....
.....[2]
 - The windscreen is designed to withstand an impact force of $5 \times 10^3 \text{ N}$. Will it break or not?
.....[1]

Specific Heat Capacity

1. The specific heat capacity of aluminium is $883 \text{ J/kg}^\circ\text{C}$. An electrical heater is used to give energy to a 1 kg block of aluminium. The results are shown in the table below:

Time (s)	0	60	120	180	240	300	360
Tempertaure ($^\circ\text{C}$)	20	20	21	23	25	27	29

- a) Plot a graph of temperature (y - axis) against time. [3]

- b) Why did the thermometer reading not increase immediately the heater was switched on?

.....[1]

- c) Use your graph to find out how long it took for the temperature to increase from 23°C to 27°C

.....[1]

- d) Calculate the heat energy required for this temperature rise

.....[1]

- e) Calculate the power of the heater

.....[1]

2. The energy needed to change the temperature of a substance depends on the mass of the substance, the temperature and the specific heat capacity of the substance.

- a) A 4 kg lump of iron is given 20 kJ of energy. It rises in temperature by 10°C . Calculate the specific heat capacity of iron.

.....[2]

- b) At the seaside on a hot day, the sand feels warm, but the sea is cold. Explain this.

.....[2]

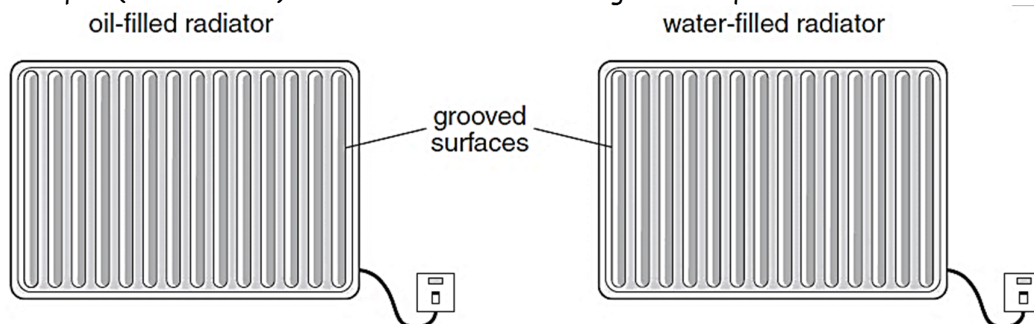
3. 200 kJ of heat is supplied to a glass cup of 2 kg . Currently, the temperature of the cup is 25°C . What will the final temperature of the cup be after heat is supplied to it? (SHC of glass = $670 \text{ J/kg}^\circ\text{C}$)

.....[3]

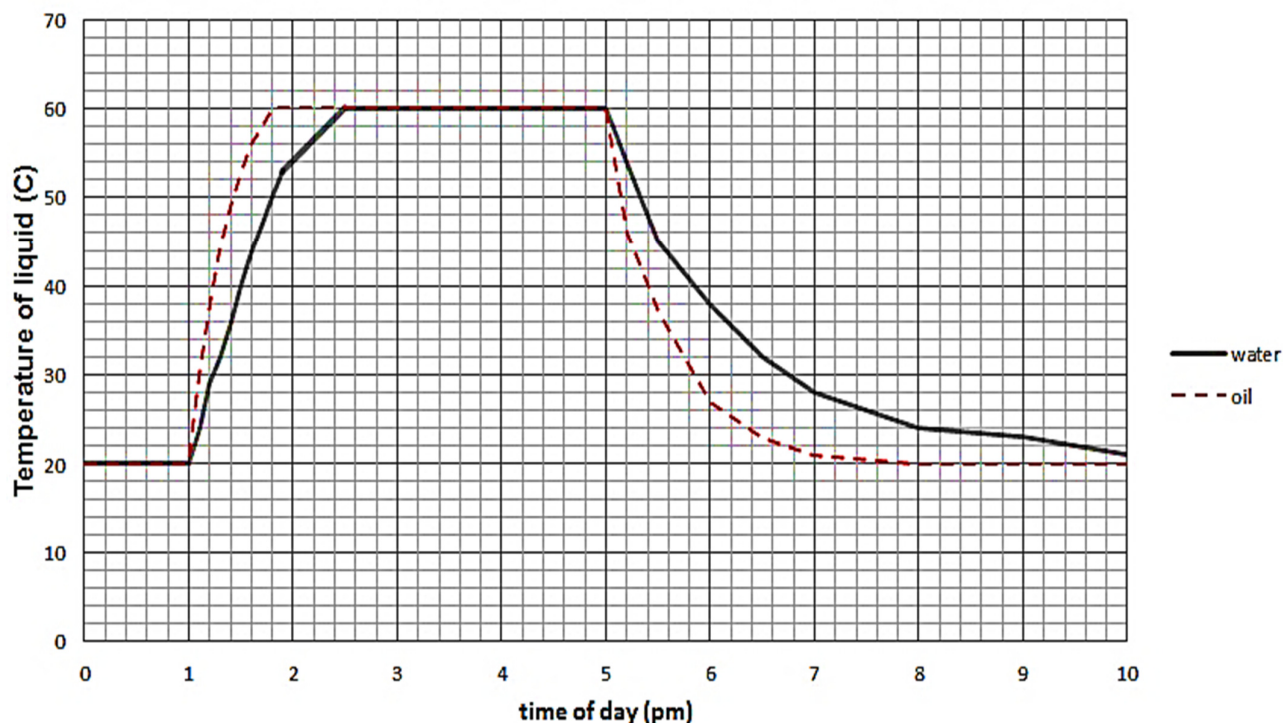
4. A heater supplies 2 kW of power to a kettle containing water of mass 4.2 kg . How long must the heater be connected to the kettle to raise the temperature of the water from 20°C to 80°C ? (specific heat capacity of water = $4200 \text{ J/kg}^\circ\text{C}$)

.....[3]

5. A room contains two electrically heated radiators. The radiators are exactly the same but contain different liquids (oil and water). Both radiators contain 10kg of the liquid.



The radiators are switched on at 1pm and left on for 4 hours. The graph shows how the temperature of the liquid in each radiator changes. The radiators' power is 278W



- a) From the graph, which substance has the larger specific capacity? How did you come to this conclusion?

.....[2]

- b) Calculate the energy that needs to be transferred to raise the temperature of the water in the radiator by 20 °C.

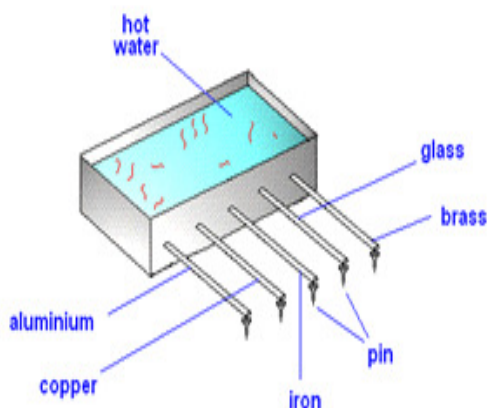
.....[2]

- c) Calculate the specific heat capacity of oil and water.

.....[3]

Conduction

1. The diagram shows a typical experiment in a school lab to show conduction



- a) Which pin do you expect to fall off first? Explain.

.....
[2]

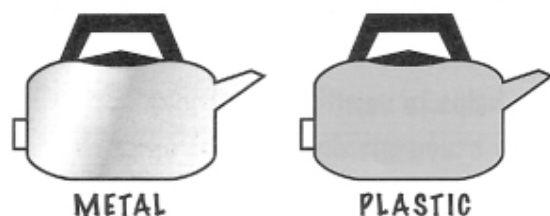
- b) Explain how heat is conducted through solids.

.....
[2]

- c) Which is a better conductor - gas, solid or liquid? Explain

.....
[2]

2. The drawing shows two kettles. One made of plastic and the other of metal



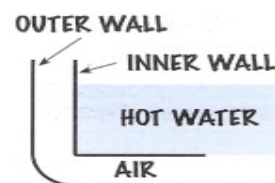
- a) If the same amount of water was boiled and then left to cool. Which would you expect to stay hottest the longest?

.....

[2]

- b) Some kettles have double walls. Suggest with reasons advantages of having a double-walled kettle

.....
[2]



3. Kurt has some ice in a cup. He wants to melt the ice as quickly as possible so he decides wrap it in a woollen cloth. Will this help? Explain your answer

.....
[2]

4. Explain the following statements:

- a) A fur coat is warmer if worn inside out

.....
[2]

- b) Two thin blankets are usually warmer than one thick one.

.....
[2]

- c) Newspapers can keep ice cream cool and chips hot

.....
[2]

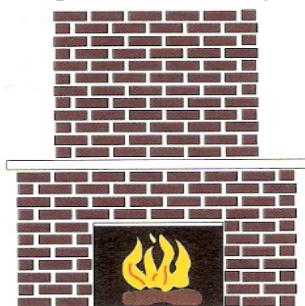
Convection

1. When an electric kettle is switched on it will take a few minutes to boil the water. Explain in detail and using diagrams how the kettle element heats all the water

.....

[2]

2. The diagram shows a fireplace:



- a) Explain why a lot of the heat is lost up the chimney

.....
[2]

- b) Explain how the fireplace helps to ventilate the room

.....
[2]

3. Describe using diagrams why we get land breezes and sea breezes. Also state when they occur.

.....[2]

.....[2]

4. Explain the following:

- a) Why you would crawl close to the floor in a smoke-filled room?

.....
[2]

- b) Why is the element at the bottom in an electric kettle?

.....
[2]

- c) Why is the freezer compartment at the top of the refrigerator?

.....
[2]

- d) How can a 500kg glider stay up in the air?

.....
[2]

Heat Radiation

1. Explain the following:

a) A car radiator is dull black

.....
[2]

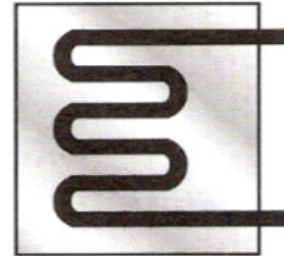
b) A solar panel works by heating up water pipes. Explain why:

i) The pipes are dull black

.....
[2]

ii) The back of the panel is shiny

.....
[2]



c) The roofs in Malta are usually painted silver. Why does this help:

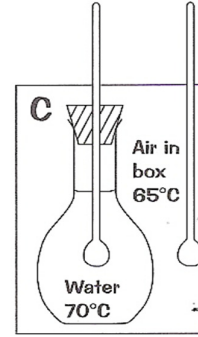
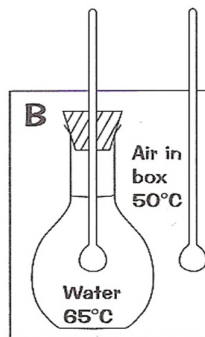
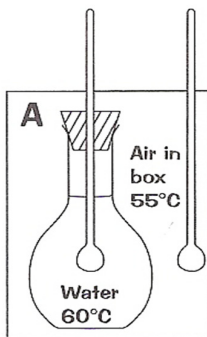
i) To keep the house cool in summer?

.....
[2]

ii) To keep the house warm in winter?

.....
[2]

2. Three flasks each containing 100ml of water are placed in closed boxes. The water in the flasks and the air in the boxes are at different temperatures as shown:



Which flask will cool the fastest? Explain your answer.

.....
[2]

3. Carbon dioxide and water vapour are greenhouse gases.

a) Explain the 'greenhouse effect' and how this is affecting our climate.

.....
[2]

b) Explain how we can reduce this problem

.....
[2]

Keeping Warm

1. Below is a list of methods of keeping heat within a house.
- Describe how each saves heat energy
 - Give the type of heat transfer that the insulation method affects.

i) Curtains[2]	ii) Double glazing windows[2]	iii) Thermostats[2]
iv) Carpets[2]	v) Wall insulation[2]	vi) Silver painted roofs[2]

2. Sharon has booked a winter holiday in a *log* cabin. She thinks that wood is a good substance to keep warm.
- Do you agree? Why?

.....
.....[2]

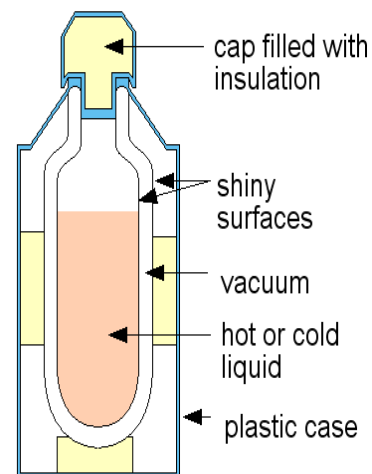
- Sarah and her husband have a relaxing evening inside while it is cold and stormy outside. Sharon touches the door, which is warm but the brass handle feels very cold. Explain why.

.....
.....[2]

3. The vacuum flask (thermos) has a number of features which help it to insulate its contents. Some are listed below. For each of them:

- Say which method of heat transfer they are reducing
- How they do this

i) There is a cap[2]	ii) Cap is filled with cork[2]
iii) The bottle is double layered[2]	iv) The shiny surfaces[2]
v) Vacuum between the two walls[2]	vi) The plastic case[2]



Static Electricity

1. Fill in the gaps with the words below:

electrons	positive	static	friction	insulating	Negative
..... electricity can build up when two materials are rubbed together. The moves from one material onto the other. This leaves a charge on one of the materials and a charge on the other [6]					

2. Lina combed her hair. As she moved the comb away from her hair, some hairs were attracted to the comb.
a) What happened to the comb to make it attract hairs?

.....[1]

- b) If the comb is held above small pieces of paper, what will Lina notice? Why?

.....

.....[1]

3. Joe carried out an experiment. He charged a Perspex sphere by rubbing it with a cloth. He then charged a polythene sphere by rubbing it with a piece of fur.

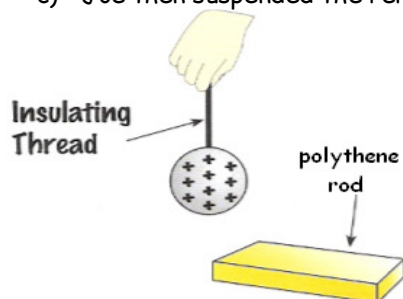
- a) What charge did the Perspex sphere and cloth gain?

.....[1]

- b) What charge did the polythene sphere and fur gain?

.....[1]

- c) Joe then suspended the Perspex sphere as shown. He held the insulating thread at all times



- i) Explain why Joe held the insulating thread at all times

.....[1]

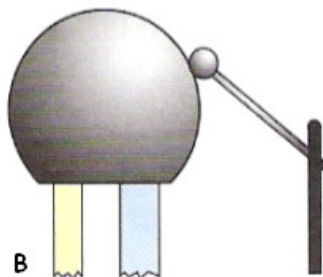
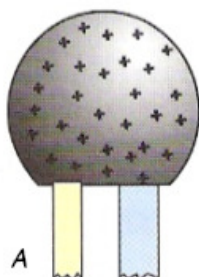
- ii) State what would happen to the sphere if a polythene rod was brought near the sphere

.....[1]

- iii) State what would happen to the sphere if the fur was brought near it.

.....[1]

4. Diagram (A) shows a metal dome of a Van de Graaf generator. The dome is charged so that it gains a positive charge. The dome is then earthed in (B)



- a) Explain (in terms of electrons & protons) how the dome discharges. (loses charge)

.....

.....

.....

.....[2]

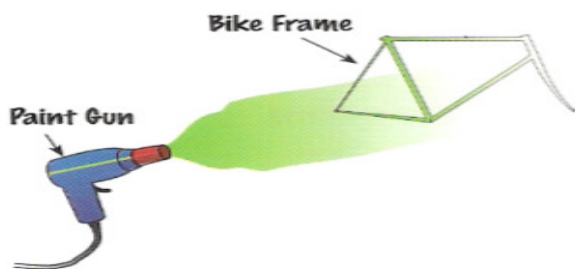
- b) The dome was then given a negative charge. It was again earthed. Explain how the dome discharges.

.....

.....[2]

Uses & Dangers of Static Electricity

1. Bicycles can be painted using electrostatic spray paint.



- a) Explain how electrostatic spray painting works.

.....

[3]

- b) Name 3 advantages of using this method

.....
[3]

- c) Describe with the help of diagrams how an electrostatic smoke precipitator works

.....

[3]

2. When Harry gets out of his car on a **dry** summer's day he gets a shock as soon as he touches the car.

- a) Explain why

.....
[2]

- b) Micha suggests that Harry gets a piece of conducting material as shown. Explain how it will prevent Harry from getting a shock

.....
[3]



3. Match up the phrases to describe what happens in a thunderstorm:

Write out your COMPLETE sentences below in the correct order

If the voltage gets big enough...

... the voltage gets higher and higher

The bottoms of the clouds become negatively charged.

...and electrons move between them

As the charge increases...

...there is a huge spark (a flash of lightning)

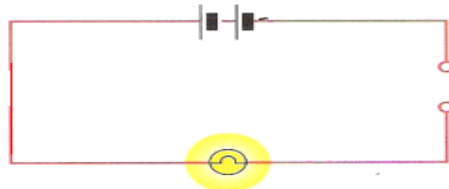
Raindrops and ice bump together...

... because they gain extra electrons

1.
 2.
 3.
 4.

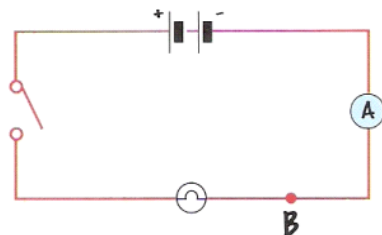
Charge, Current, Voltage & Resistance

1. This is an electrical circuit



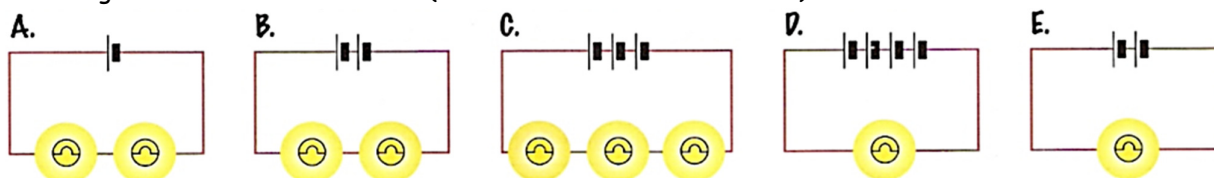
- Mark on the diagram the direction of conventional current. Mark it as ' I ' [1]
- Show on the diagram the flow of electrons. Mark as ' e ' [1]
- On the diagram draw an ammeter and a voltmeter [2]

1. In the circuit below the cells provide a potential difference of 9V. A current of 0.2A flows for 5 minutes.



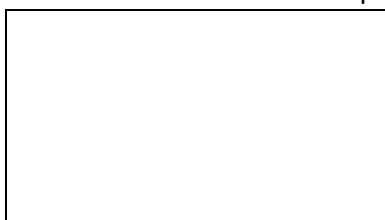
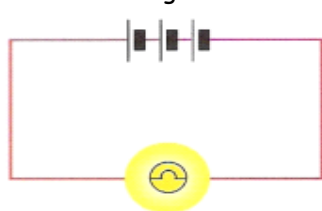
- Calculate the charge that passes through point B in these 5 minutes.
.....[1]
- Calculate the energy transferred in the circuit in these 5 minutes.
.....[1]

2. The diagram shows several circuits (each cell and bulb are identical)



- In which circuit is the lamp the brightest? Explain
..... [2]
- In which circuit does the largest current flow? Explain
..... [2]
- Which circuit has the greatest resistance? Explain
..... [2]
- Which circuit has the greatest potential difference? Explain
..... [2]

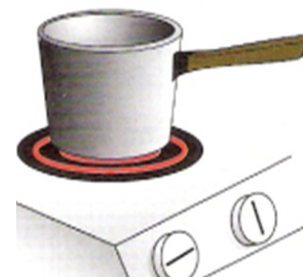
3. The circuit diagram shows three batteries in series with a lamp.



- Redraw the circuit and add an ammeter [1]
- What does an ammeter measure?
.....[1]

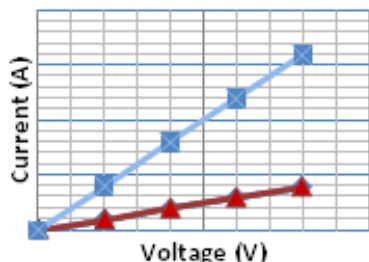
4. An electric current passing through an electrical device has a heating effect.

- Name three devices which make use of this.
.....[3]
- The diagram shows a 'ring' on an electric cooker. Explain why the ring gets hot when a current passes through it.
.....
..... [2]



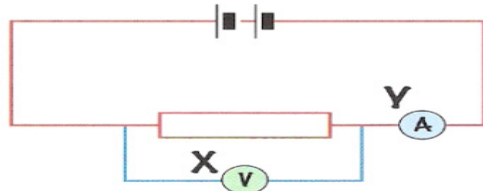
Resistance & Ohm's Law

1. The diagram shows a current-voltage graph:



- Which line, A or B could show a longer wire? Explain[2]
- Which line, A or B could show a thinner wire? Explain.[2]
- Which line, A or B could show a higher temperature? Explain[2]
- Which line, A or B could show a copper and which a nichrome wire? Explain[2]

2. The circuit below was used to measure the resistance of a fixed wire



- Name the components X and Y[2]
- If the first voltage is 6V and current is 1.5A
 - Calculate the resistance of the resistor[1]

ii) If the voltage is increased to 9V what would the current flowing through the resistor be?

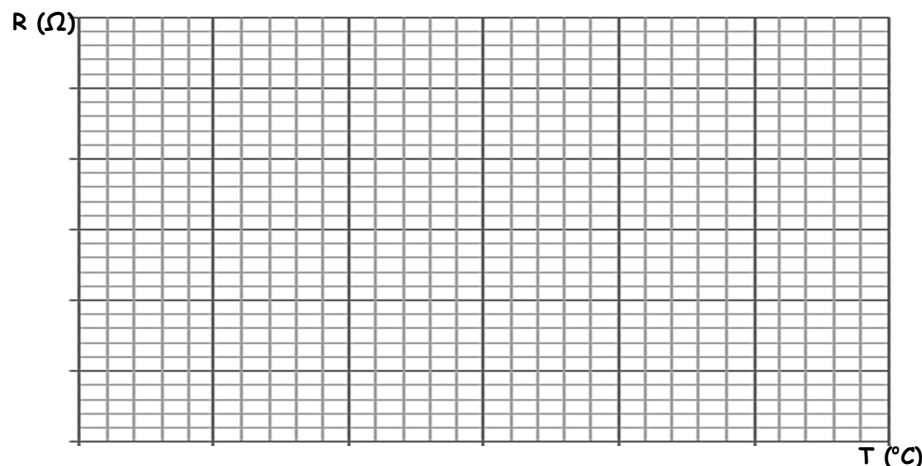
.....[1]

iii) A **different** resistor of 30Ω was placed in the circuit. Calculate the voltage if the current is 0.2A

.....[1]

3. The resistance of a thermistor was found over a range of results shown below:

Resistance (Ω)	3000	1800	1250	900	600	500	350	300	300
Temperature ($^{\circ}\text{C}$)	0	10	20	30	40	50	60	70	80



- Plot the graph [3]
- Describe what happens to the resistance of the thermistor as the temperature changes.[2]

c) If the voltage across the thermistor is 12V. Calculate the current at a temperature of 40°C

.....[1]

d) If the voltage is 12V and the current is 0.0096A

i) Calculate the resistance of the thermistor

.....[1]

ii) At what temperature were these readings taken?[1]

Other Devices (Current – Voltage Graphs)

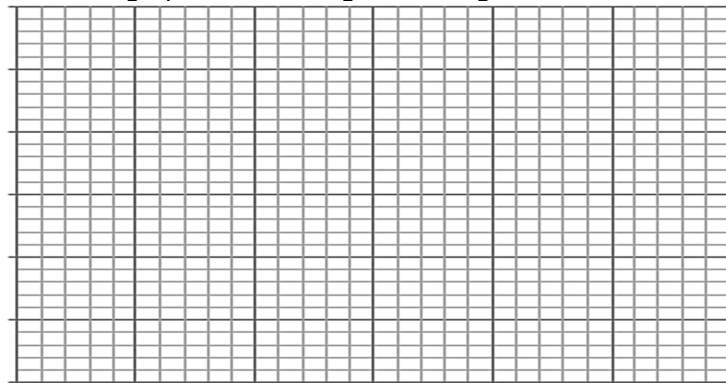
1. Jacob decides to test how the current through a filament lamp changes as the voltage across the lamp is changed.

- a) Draw a circuit diagram to show how he should set up the circuit [2]

The results obtained are shown below:

Voltage (V)	0.0	2.0	4.0	6.0	8.0	10.0	12.0
Current (A)	0.0	1.0	1.4	1.7	1.9	2.1	2.2

- b) Plot a graph of current against voltage



- c) Use your graph to calculate:

- i) The resistance of the lamp when the voltage is 3V

.....

..... [2]

- ii) The resistance of the lamp when the voltage is 9V

.....

..... [2]

- d) What happens to the resistance of the lamp if the current passing through increases? Explain

.....

..... [2]

2. Sketch I-V graphs for the following components:

Resistor

Thermistor

Diode

[6]

3. Use the data in the table to plot a graph of resistance vs. light intensity.

Resistance (k Ω)	100	55	40	20	1	0.1
Light intensity	0.5	2	3	5	7.5	10

- a) How does the resistance change as the light gets brighter?

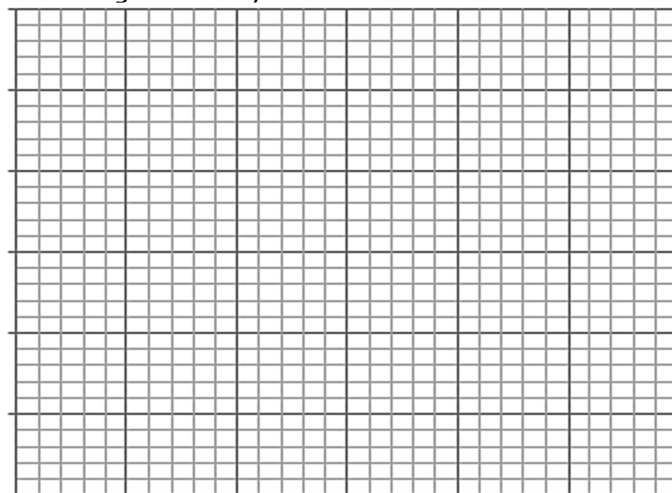
..... [1]

- b) Looking at the slope of the graph, describe how the resistance changes in bright light compared to dim light, as the light gets brighter

.....

.....

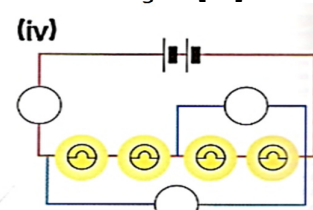
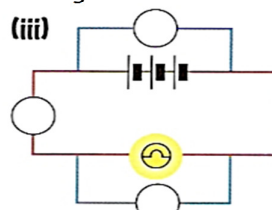
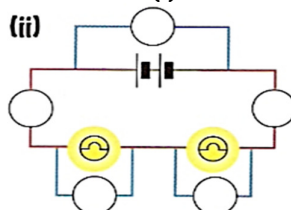
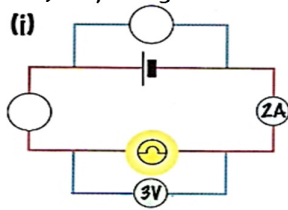
..... [1]



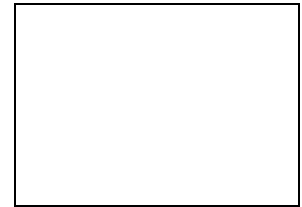
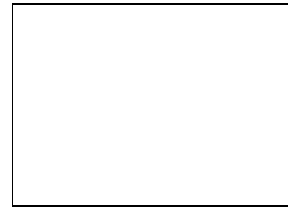
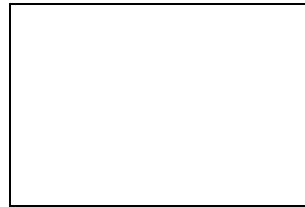
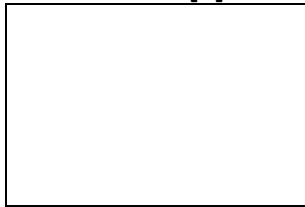
Series Circuits

1. In the series circuits below all the components are identical.

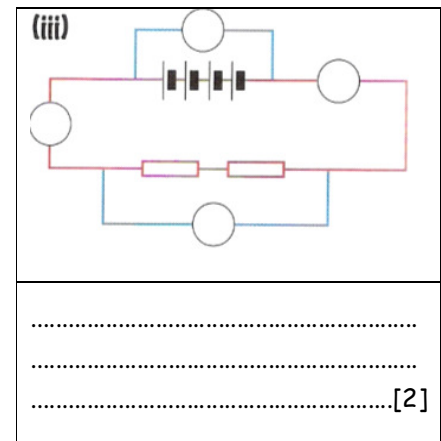
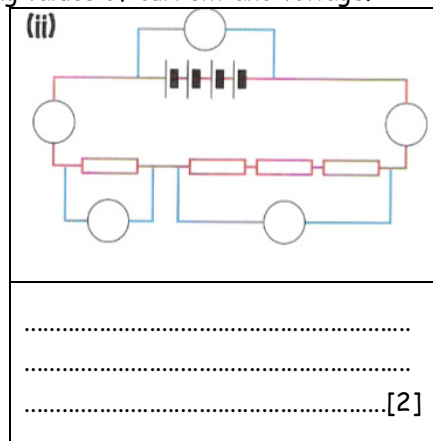
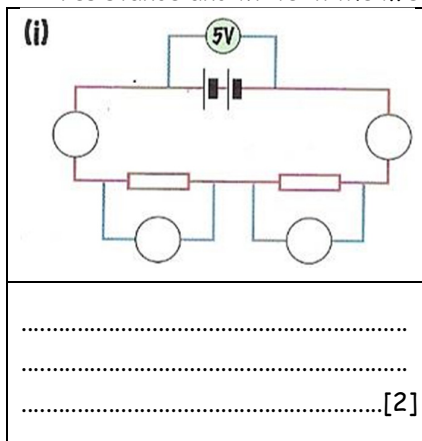
a) By using the information in circuit (i) write in the missing values of the currents and voltages: [13]



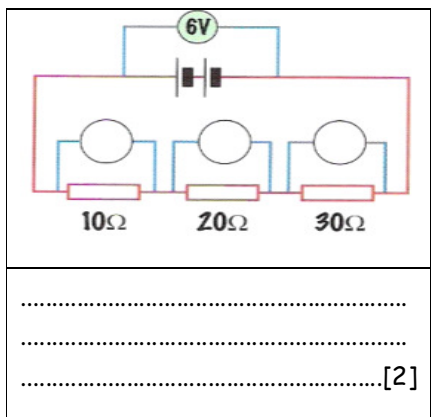
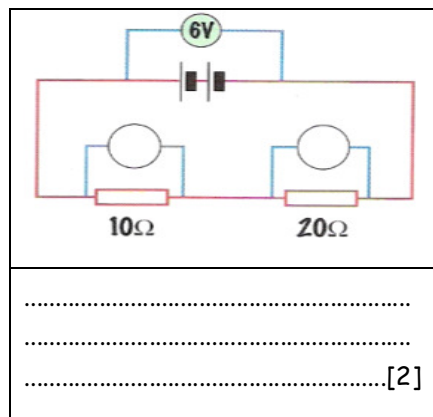
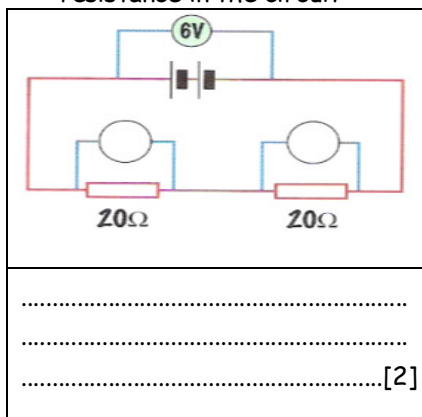
b) For each one draw a circuit diagram including ammeters and voltmeters if the bulb has twice the resistance [8]



2. In the following circuits each resistor has a resistance of 10Ω . For each circuit calculate the total resistance and write in the missing values of current and voltage.



3. For each of the following circuits write in the voltage across each resistor and calculate the total resistance in the circuit



4. Match up these descriptions

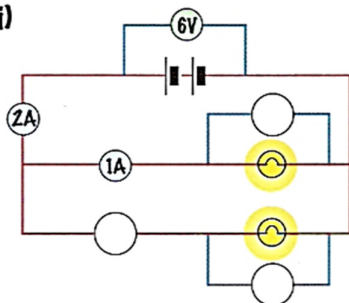
A	Same everywhere in the circuit
B	Shared out between the components
C	Can be different for each component

	Voltage (potential difference)
	Current
	TOTAL voltage

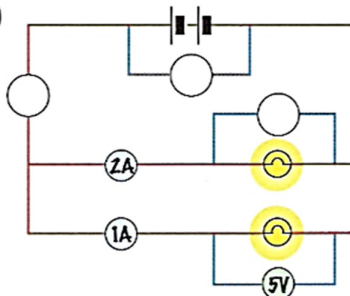
Parallel Circuits

1. For the following circuits write in the missing values of current and voltage. [10]

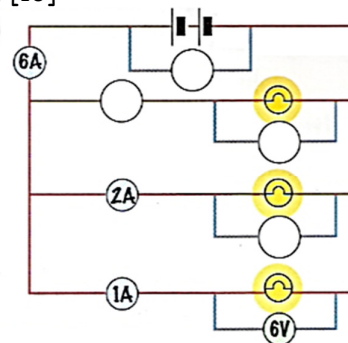
(i)



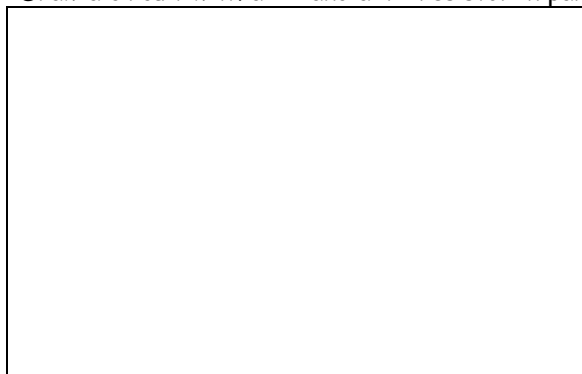
(ii)



(iii)



2. Draw a circuit with a 2Ω and a 4Ω resistor in parallel, with a 6V battery [2]



- a) What is the current in the 2Ω resistor?

.....[1]

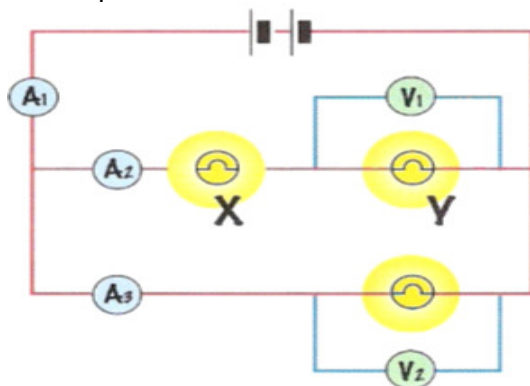
- b) What is the current in the 4Ω resistor?

.....[1]

- c) What is the current through the battery?

.....[1]

3. The lamps in the circuit shown are all identical. The voltage supply is 6V.



- a) What is the voltage V_1 ?[1]

- b) What is the voltage V_2 ?[1]

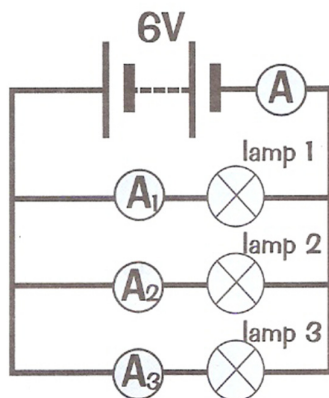
- c) If A_1 reads 6A, what is the reading on:

i) A_2 [1]

ii) A_3 [1]

- d) If the resistance of each lamp is 20Ω , what is the total resistance of X and Y?[1]

4. Complete the following with: *less, series, branch, parallel, A_2 and A_3 , A_1 , more, same*



If the lamps are connected in _____, the current in the main part of the circuit splits up and goes through each _____. The voltage across each lamp is the _____. So the brightness of the lamps stays the same the _____ lamps you add in _____.

But each lamp will be dimmer if you add lamps in _____ because the voltage is divided for components in _____.

The ammeter reading at _____ is lower than at A but is the same as those at _____. [9]

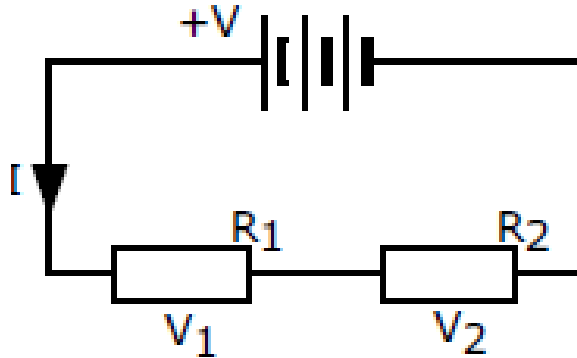
Circuit Training

For each of the circuits shown in the diagram 1,2,3 and 4 carry out the necessary calculations to find the voltage, current and resistance

Each row represents a different circuit with different values.

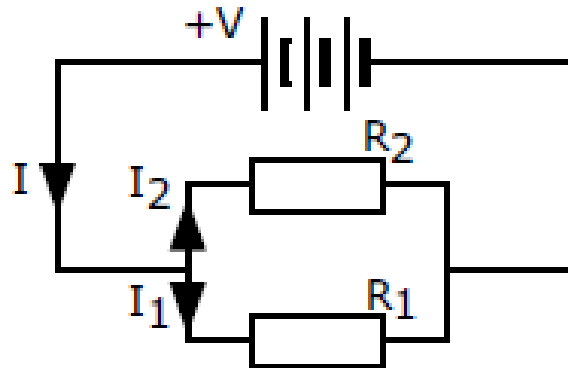
Show ALL working on a separate A4 paper

1.



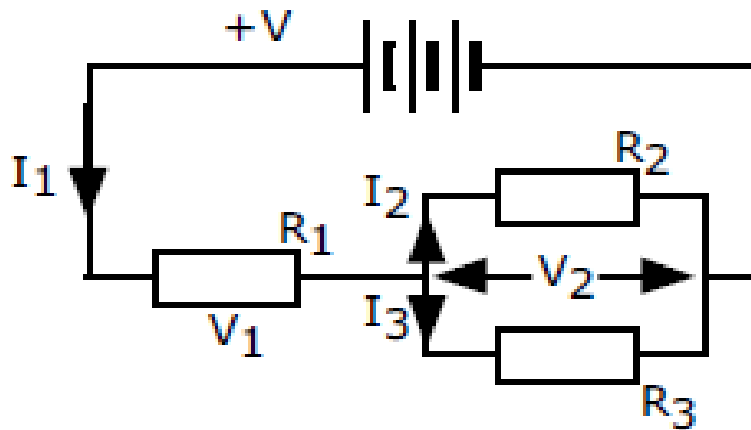
	V	I	R_1	R_2	R_{TOTAL}	V_1	V_2
A			3Ω	2Ω			
B		2A	5Ω		6Ω		
C		0.4A				1V	0.6V
D	11V		900Ω	200Ω			2V
E			0.05A		$13k\Omega$	150V	500V

2.



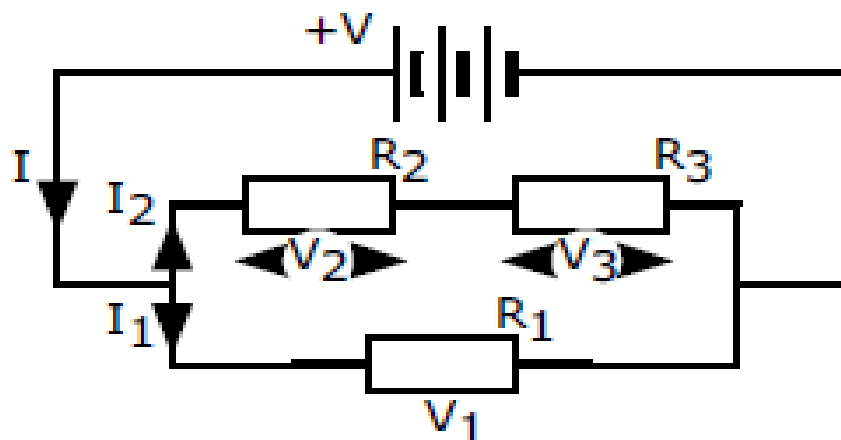
	V	I	R_1	R_2	R_{TOTAL}	I_1	I_2
A	15V		5Ω	3Ω			
B		10mA		$4k\Omega$	$0.8k\Omega$		2mA
C			2.4Ω		2Ω	625mA	125mA
D	4V	0.31A		64Ω		0.25A	
E			100Ω	2Ω		0.09A	4.5A
F					2.1Ω	1A	2.3A

3.



	V	I_1	R_1	R_2	R_3	R_{TOTAL}	I_2	I_3	V_1
A	11V		2Ω	4Ω	6Ω				
B	1.5V	50mA		60Ω	20Ω	30Ω			
C				100Ω	300Ω	275Ω	0.06A	0.02A	
D	20V		$6k\Omega$		$5k\Omega$	$10k\Omega$			
E			1Ω	5Ω		3Ω			3V

4.



	V	I	R_1	R_2	R_3	R_{TOTAL}	I_1	I_2	V_1	V_2	V_3
A	1.5V		3Ω	2Ω	1Ω						
B		2A	5Ω		30Ω	4.5Ω					
C		0.2A	100Ω			87Ω				5V	12.5V
D							10mA	2mA		2.4V	0.6V
E			2Ω	1Ω		1.8Ω	13.5A				
F						0.99Ω		0.02A		0.02V	1.98V

Paying for Electricity

1. The table below shows the power rating and how long they are on each week. Complete the table: [12]

Appliance	Power Rating (W)	Time used each week	Energy Transferred (J)	Energy Transferred (kWh)	Cost per week (c) (20c per unit)
Kettle	1500	6h			
Geyser	3000	5h			
Cooker	4000	15h			
Hair Drier	750	3h			
Iron	1000	2h			
Washing Machine	2500	6h 30min			
Tumble Drier	2000	30min			
Television	300	3h 45min			

2. The diagram shows part of an electricity bill:

SAVE ENERGY LTD.

Customer No. 3 263 1319 12	Statement Date 26 Sept 2009
Most recent reading	47209.4 kWh
Previous reading	45959.5 kWh
Units used	kWh (Units)
units at 22c per unit	=

- a) Calculate the number of units used

.....
[1]

- b) Calculate the cost of electricity

.....
[1]

3. Lisa's dad is always complaining that when she leaves her room, she always leaves all the electrical appliances on. Lisa writes down the power rating of the appliances: three light bulbs (100W each), one CD player (95W), The computer system (290W), electrical heater (2.4kW)

- a) Calculate the total energy being used each night if she leaves them on for 5 hours.

.....

[2]

- b) Calculate the cost of leaving these appliances for one night (cost is 20c per unit)

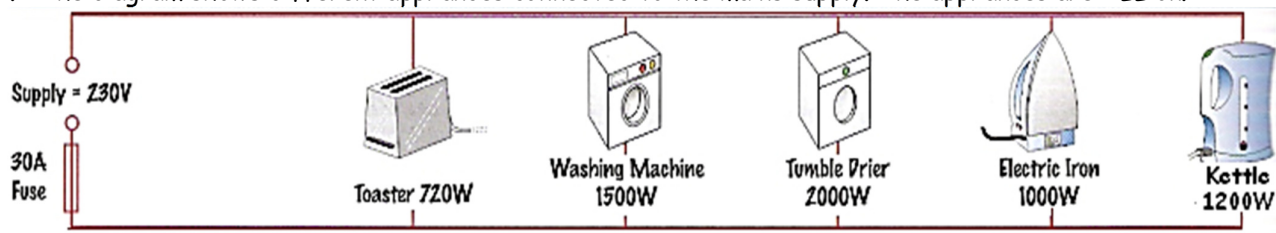
.....
[2]

- c) What would the cost be over a year if she did this every night? (1 year = 365 days)

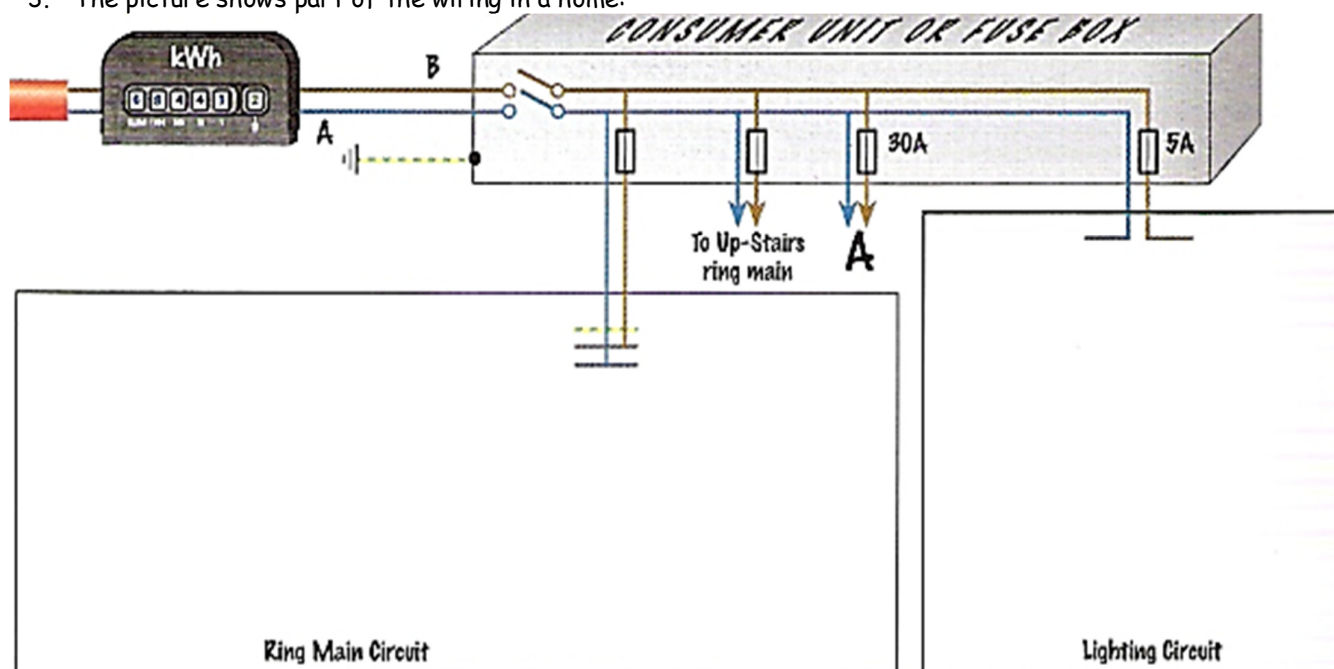
.....
[2]

Domestic Electricity

- Explain the difference between A.C. and D.C.
.....[2]
- The diagram shows different appliances connected to the mains supply. The appliances are ALL on.



- Calculate the current flowing through each appliance
.....
.....[2]
 - Is the fuse big enough to take all of these appliances when they are on? Explain
.....
.....[2]
- The picture shows part of the wiring in a home:



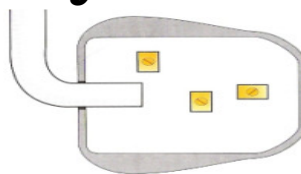
- Complete the diagram by
 - Drawing a circuit diagram for the mains circuit with 4 sockets in it [2]
 - Drawing a circuit for the lighting circuit with 3 light sockets [2]
- How many 100W light bulbs can be connected in the lighting circuit? The supply voltage is 240V
.....
.....[2]

Electrical Safety – Fuses

1. Complete the diagram by:

a) Drawing the missing parts [2]

b) Labelling ALL the parts [2]



2. It is important the correct fuse is fitted with an appliance. Complete the table:

Appliance	Power Rating (W)	p.d. (V)	Working Current (A)	Fuse Size (1A, 3A, 5A, 13A)
Iron	1000	240		
Food Mixer	460	240		
Hi-Fi	115	240		
Hoover	1500	240		
Computer	100	240		
Hair Drier	500	240		

3. The diagram shows a hair drier with a plastic casing

a) The hair drier has two wires. Which are they?

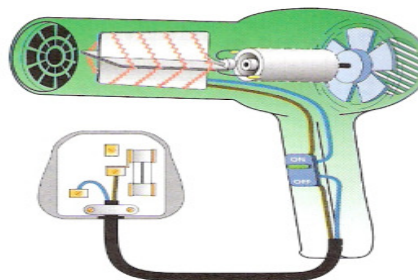
.....[1]

b) Which wire is missing?[1]

c) Explain why the hair drier is safe without this wire

.....

.....[2]



d) The diagram shows the information on the hair drier. Name a suitable fuse that can be used. (show all working)

.....[2]

e) During use a fault occurs and a current of 30A flows through the live wire

i) Suggest a possible reason for a fault occurring[2]

ii) Explain what happens to the fuse and how it protects the hair drier

.....

.....[2]

4. The voltage across each lamp is 12V. Each has a resistance of 5Ω

a) Calculate the current passing through each lamp

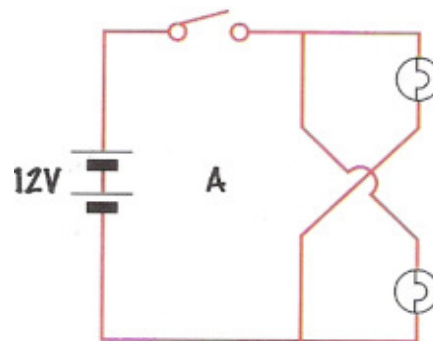
.....[2]

b) Suggest a suitable fuse for the circuit

.....[1]

c) Calculate the TOTAL power supplied to the lamps

.....[2]



Electrical Safety – Earthing & Circuit Breakers

1. An electric cooker with a metal case has three wires connected to the plug.

a) What are the names of the three wires and their colours?

.....
[2]

b) Which wire should not have a current pass through it under normal circumstances?

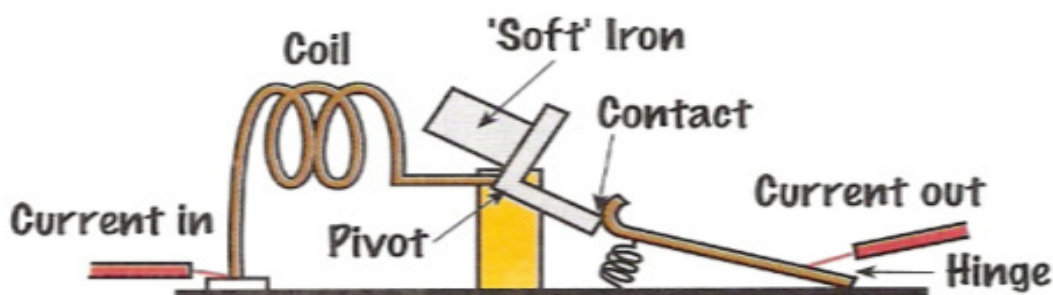
.....[2]

c) To what part of the cooker should this wire be connected?.....[2]

d) If a fault develops in the cooker, explain how the user is protected from an electric shock.

.....
[2]

2. The diagram shows a circuit breaker:



a) What is the job of the circuit breaker?

.....[1]

b) What advantage do circuit breakers have over fuses?

.....[1]

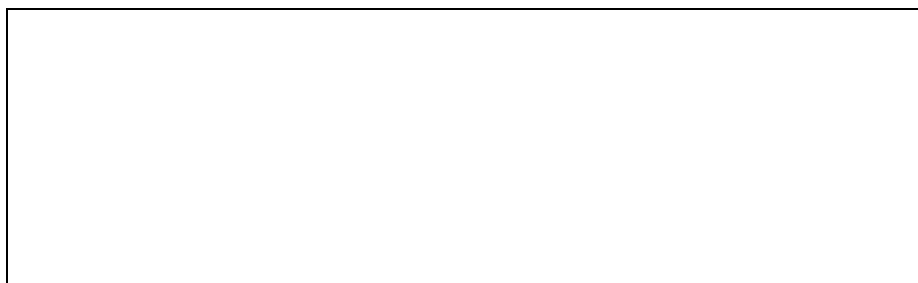
c) i) Explain in detail, what would happen if a high surge of current were to flow through the circuit breaker?

.....
[2]

ii) How does this make the circuit safe?

.....[2]

iii) Draw a diagram of the circuit breaker after a high surge of current

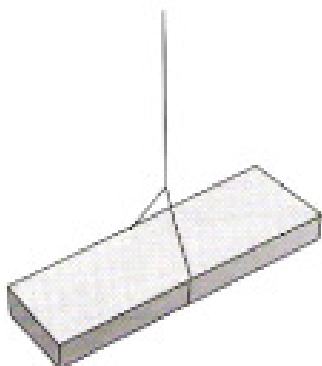


d) Why should fuses and circuit breakers ALWAYS be put on the live wire?

.....[2]

Magnetism

1. The diagram shows a bar magnet which is allowed to swing freely in the Earth's magnetic field.



- a) Complete the diagram by including the direction of the Earth's magnetic field and labeling the poles on the magnet [2]

- b) Another piece of metal is brought up to the bar magnet.

- i) What can you say about the metal if it is attracted to the magnet?
.....[1]

- ii) How could you test if the metal is a magnet or not? Explain.
.....
.....[1]

- c) Complete the diagrams by drawing the shapes and directions of the magnetic fields:

(i)



(ii)



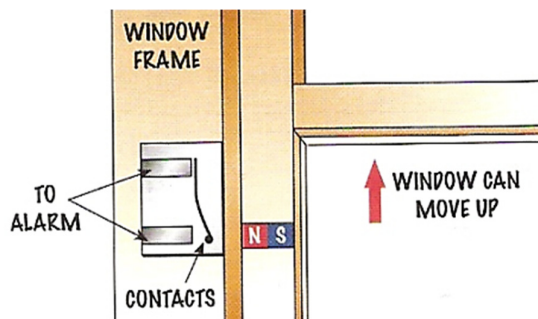
(iii)



(iv)



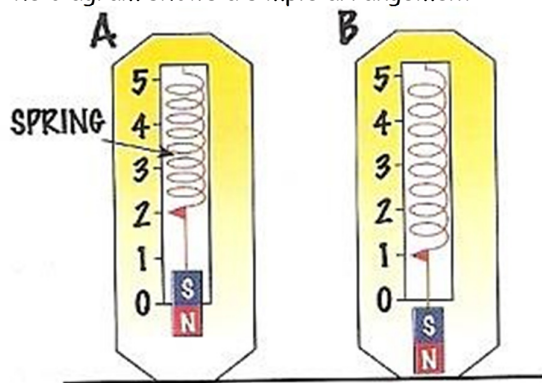
2. The diagram shows a magnetic switch. It is being used as a burglar alarm,



Explain how the alarm works.

.....
.....
.....
.....[3]

3. The thickness of paint on cars can be measured by using a magnet attached to a Newton meter. The diagram shows a simple arrangement



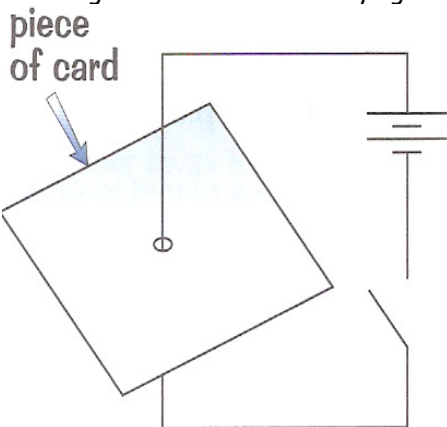
- a) In which diagram is the paint the thickest?

.....[3]

- b) Explain how this paint thickness measurer works.
.....
.....
.....
.....[3]

Electromagnetism

1. The diagram shows a wire carrying a current passing through a piece of flat cardboard:



- a) Some iron filings are sprinkled onto the card. On the diagram draw the pattern that will be shown after the switch is turned on. Also show the direction and state the rule used.....[3]
- b) State True or False:
 - i) Inside a solenoid, the magnetic field is like a bar magnet's[1]
 - ii) The iron core keeps its magnetism when the current is switched off[1]
 - iii) As more turns are added to the coil, the magnetic field gets weaker[1]

2. Kurt is carrying out an investigation into the factors that affect the strength of an electromagnet. He decides to investigate how the strength depends on the number of turns of wire on the coil.

- a) State the other two factors that will affect the strength of the electromagnet

i) ii) [2]

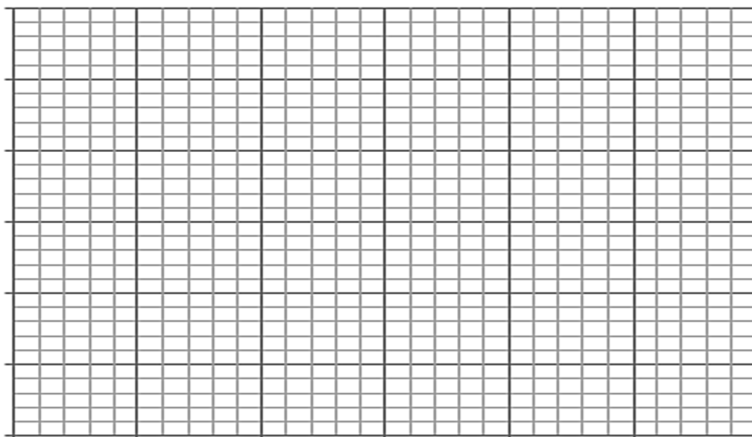
- b) Kurt decided he would see how many paper clips the electromagnet could pick up. How will he do it?

.....

[3]

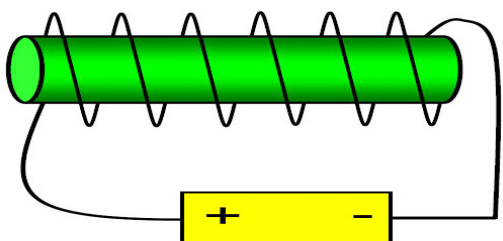
- c) The results of Kurt's experiment are shown below:

Number of turns	0	10	20	30	40	50	60
Number of paper clips	0	2	4		8	11	12



- i) Plot a graph of number of turns (x - axis) against the number of paper clips (y-axis) [4]
- ii) Find the missing value [1]
- iii) What conclusion can Kurt come to from these results?

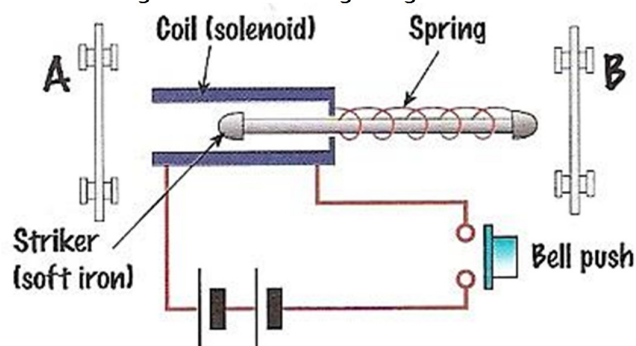
3. The diagram shows a solenoid connected to a battery



- a) Draw the shape of the magnetic field (show North) [2]
- b) What effect does the soft iron core have?[1]
- c) Describe another two ways to increase the strength of the magnetic field
 - i) ii)[2]

Uses of Electromagnetism

1. The diagram shows a 'ding-dong' bell. The bell works by hitting the chimes A and B



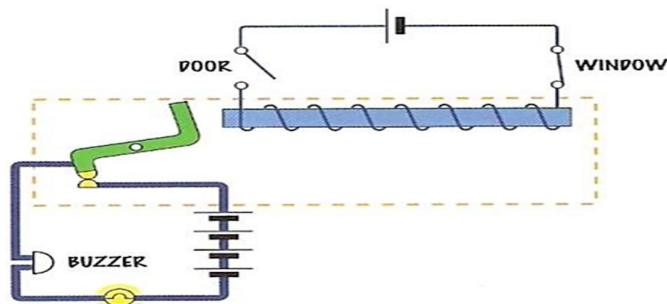
- What happens to the solenoid when the bell is pressed?
.....[1]
- Why is the striker made of soft iron?
.....[1]
- What happens to the striker when the bell push is pressed?
.....[1]
- What is the purpose of the spring?
.....[1]

- Describe how the bell works
.....
..... [2]

- The manufacturer receives a complaint that the bell is not loud enough. Suggest two ways they could change the design to make it louder. Explain your answers
.....
..... [2]

2. The diagram shows a simple arrangement for a burglar alarm. A door has been opened and the alarm has been activated

- Explain fully what happens to the alarm when the door is closed [3]
.....
.....
.....

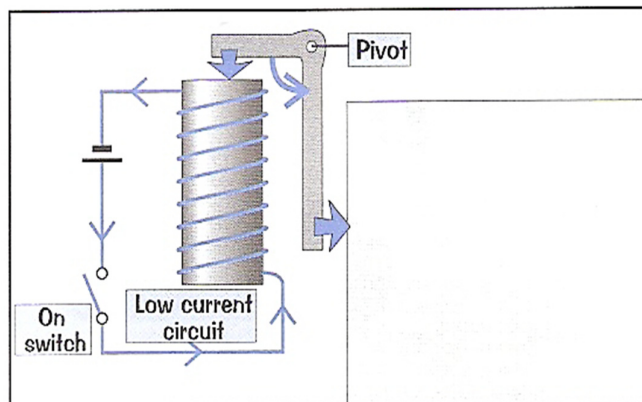


- Draw a diagram to show the alarm after the door has been closed [3]
.....
.....
.....

3. The diagram shows a starter motor relay

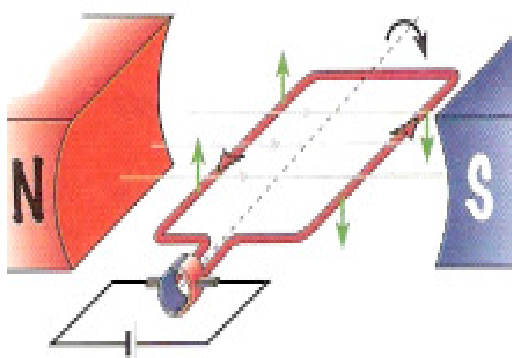
- Draw the missing part of the diagram (labels) [3]
- What part of the car electrics uses the relay switch and what is its function?
.....
..... [2]

- Explain what happens when the switch is closed
.....
.....
..... [2]



The Motor Effect

1. The diagram shows a simple direct current motor:



The following statements describe how the motor works

A ... which interacts with the magnetic field of the magnet ...

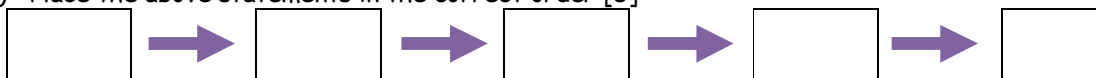
B ... so the coil rotates ...

C ... creating a force on both sides of the coil ...

D ... this creates a magnetic field ...

E ... as a current flows through the coil ...

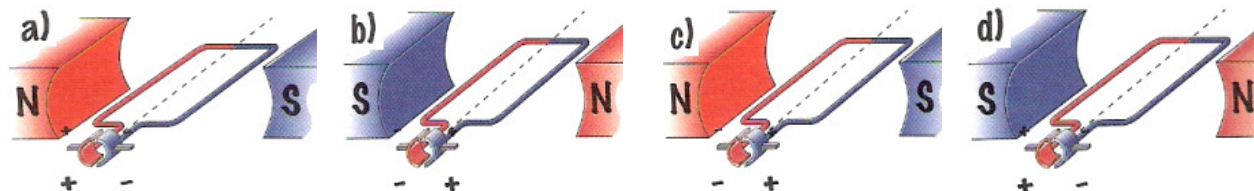
- a) Place the above statements in the correct order [5]



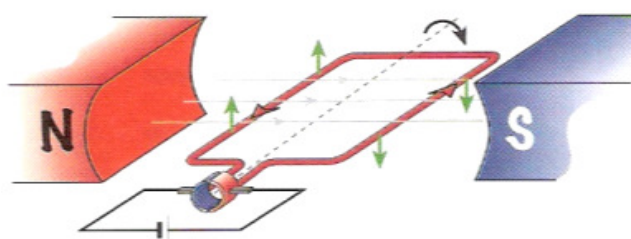
- b) For each of the following statements state what effect they will have on the speed of the motor:

- i) The strength of the magnets is decreased.....[1]
- ii) A soft iron core is placed inside the coil.....[1]
- iii) The number of turns is increased.....[1]
- iv) The size of the current through the coil is decreased.....[1]

2. For the electric motors below find the direction of rotation of the motor



3. The diagram shows an electric motor

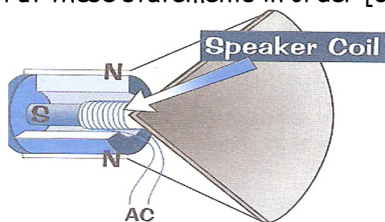


Describe how it works paying particular attention to the commutator

.....

[4]

4. Put these statements in order [3]



This creates sounds.

These make the coil move back and forth

between the poles of the magnet.

are fed into the speaker coil.

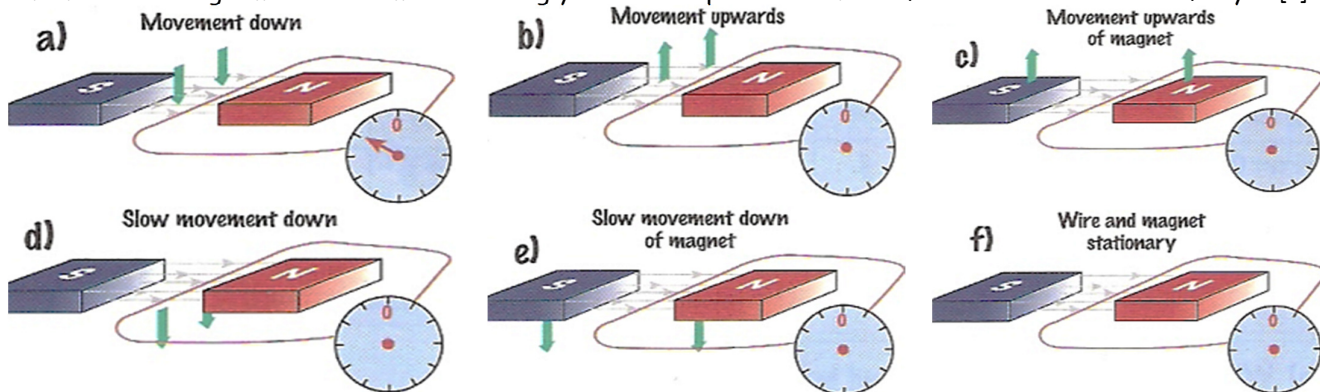
the cardboard case vibrate.

These movements make

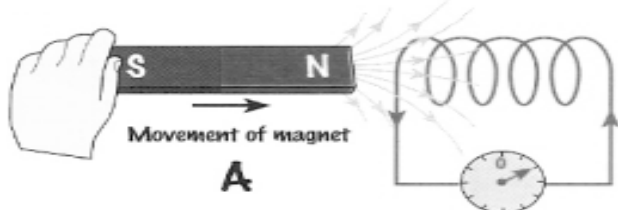
AC signals from the amplifier

Electromagnetic Induction

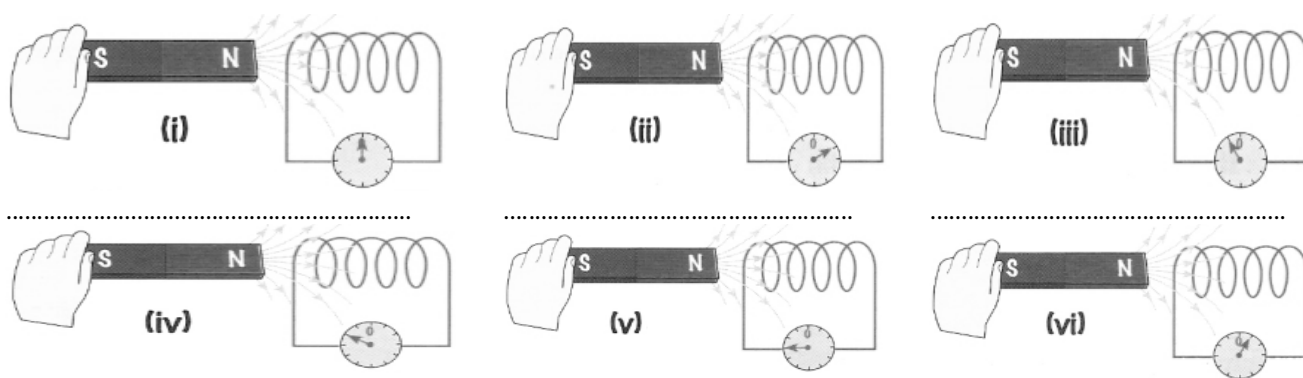
1. For each diagram draw the meter reading you would expect to see. The first one has been done for you [5]



2. The diagram shows a bar magnet moving towards a coil of wire. The meter shows a reading.

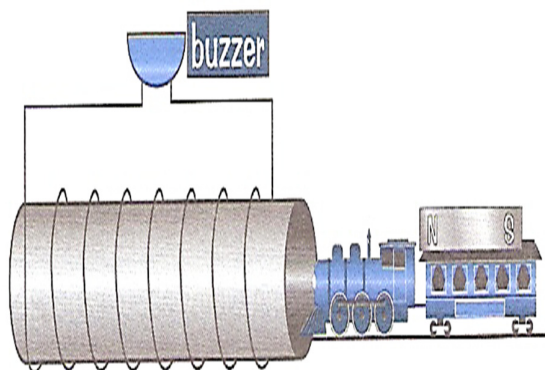


- a) For the diagrams below describe the motion of the magnet compared to diagram A. You should use the words: *'same speed'*, *'faster'*, *'slower'* same direction', *'opposite direction'*, *'no movement'*



- b) State three ways of altering the induced voltage:

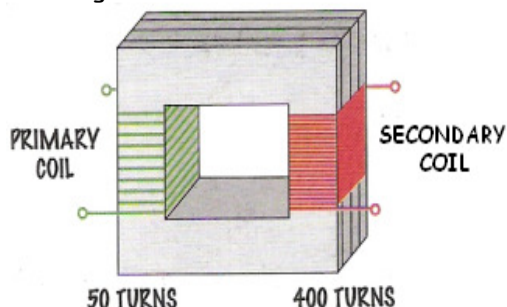
3. A model train travels at high speed into a tunnel. A bar magnet is fixed on top of the train. A coil of insulated wire is wound around the tunnel and underneath the track (connected to a buzzer)



- a) Explain why the buzzer sounds when the train passes through the tunnel
-[2]
- b) Would the buzzer sound if the train **stopped** inside the tunnel? Explain
-[2]
- c) Suggest two ways to make the buzzer sound louder (without dismantling the tunnel)
-[2]

Transformers

1. The diagram shows a transformer



- a) Name the material used to make the core of the transformer

.....[1]

- b) Explain in detail how the transformer works:

.....

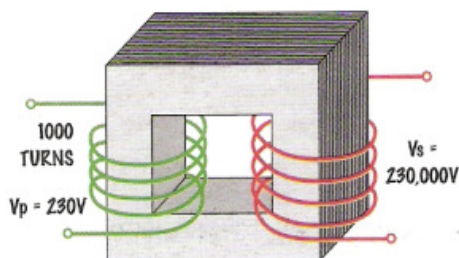
.....[3]

- c) If the voltage applied to the primary coil is 12V, calculate the voltage secondary coil

.....
 [2]

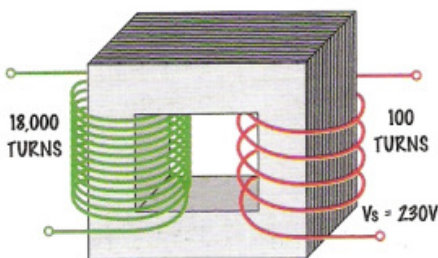
2. For each transformer calculate:

- a) The number of turns in the secondary coil



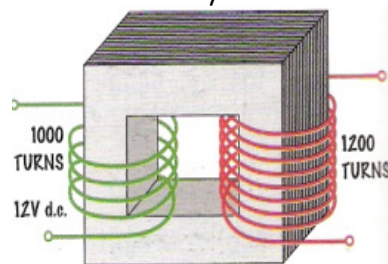
.....
[2]

- b) The voltage across the primary coil



.....[2]

- c) The voltage across the secondary coil



.....[2]

3. A transformer has 100 turns in its primary coil and 100 turns on its secondary coil. The voltage across the primary coil is 240V

- a) Calculate the voltage across the secondary coil

.....
 [2]

- b) If the current in the primary coil is 2A and the current in the secondary coil is 18A. Calculate:

- i) The power input

.....[1]

- ii) The power output

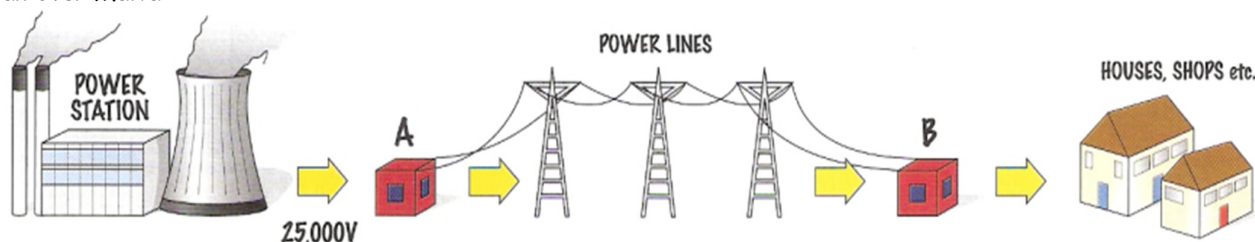
.....[1]

- iii) The efficiency of the transformer

.....
 [1]

Transmission of Electricity

1. The diagram below shows how electricity is distributed from power stations to shops, factories and houses all over Malta



- a) What is the name given to this electrical distribution

.....[1]

- b) What are transformers A and B used for?

.....[1]

- c) Name the type of transformer in A and B and explain why they are used

A:

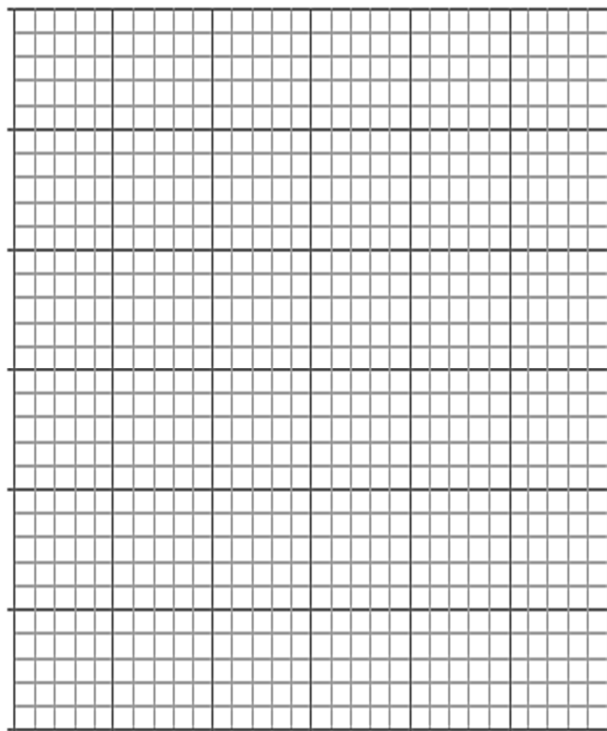
B:

.....[2]

..... [2]

2. A generator at a power station produces 500MW of electrical energy. The table below gives possible voltage and current values that the electricity can be transmitted at:

Voltage (kV)	100	150	200	250	300	350	400
Current (A)	5000	3300	2500	2000	1700	1400	1300



- a) Plot a graph of voltage against current [3]

- b) If the electricity is transmitted at a voltage of 275kV, what current flows through the power lines?

.....[1]

- c) Explain why the electricity is transmitted at a high voltage and a low current

.....

.....[1]

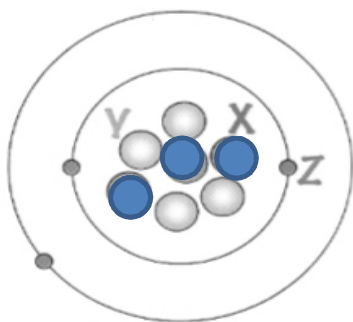
- d) Explain why it is important to consider the thickness of the cable when transmitting electricity

.....

.....[1]

Structure of the Atom

1. The diagram shows an atom of Lithium.



- a) Name each of the particles X, Y & Z [3]

X Y Z

- b) Complete the table: [3]

Atomic Particle	Relative mass	Relative charge
		+1
	1	
	$\frac{1}{2000}$	

2. Complete the following table

Element	Copper	Silicon	Magnesium	Iron	Oxygen	Lead	Gold	Carbon	Neon	Calcium
Symbol	${}^{64}_{29}\text{Cu}$	${}^{28}_{14}\text{Si}$	Mg	${}^{56}_{26}\text{Fe}$	${}^{16}_8\text{O}$	${}_{82}\text{Pb}$	${}^{197}_{79}\text{Au}$	${}^{12}_6\text{C}$	Ne	${}^{40}_{20}\text{Ca}$
Mass Number			24							
No. of Protons			12							
No. of Neutrons						125			10	
No. of Electrons									10	20

3. Two isotopes of cobalt exist. Cobalt-60 and Cobalt-59 $\rightarrow {}^{60}_{27}\text{Co}$ and ${}^{59}_{27}\text{Co}$

- a) What is an isotope?

.....
[2]

- b) For each isotope write down:

i) The number of protons: Cobalt-60 = Cobalt-59 =[2]

ii) The number of neutrons: Cobalt-60 = Cobalt-59 =[2]

iii) The number of electrons: Cobalt-60 = Cobalt-59 =[2]

4. Copy and complete the paragraph with the following words:

<i>atomic</i>	<i>mass</i>	<i>alpha</i>	<i>decay</i>	<i>neutrons</i>	<i>electrons</i>
<i>stable</i>	<i>beta</i>	<i>three</i>	<i>element</i>	<i>energy</i>	<i>protons</i>

Isotopes if the same have equal numbers of and but different numbers of Hence they have the same number. Every has at least different isotopes but usually only one or two ones. If a radioactive isotope decays, radiation is emitted. If an or a particle is emitted then a different is formed [11]

Three Types of Radiation

1. The diagram below shows alpha, beta and gamma radiation being fired at a line of obstacles



- a) For each particle draw a line to show the path it travels before it is absorbed [3]
 b) Give a reason why alpha particles only penetrate a short distance

.....[1]

2. Here is a list of properties of alpha, beta and gamma radiation. Match them to the correct radiation: [11]

Alpha				
Beta				
Gamma				

Electromagnetic Waves

Electrons

Very short wavelength

Mass no. = 4

Charge = -1

Charge = +2

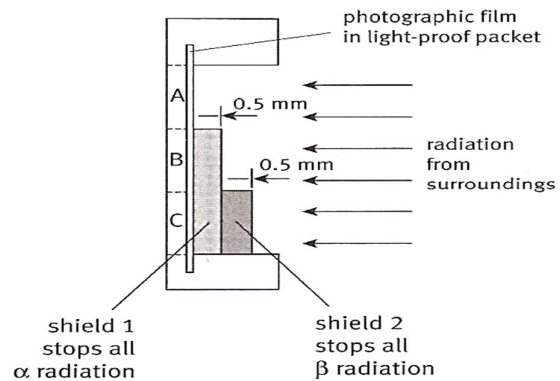
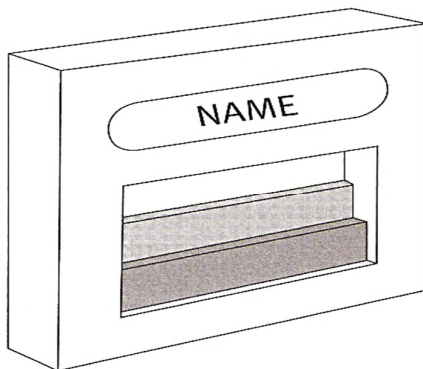
Zero mass

 γ β α

Helium nucleus

[11]

3. The picture on the left shows a type of radiation badge that is worn by radiation workers. The diagram on the right is a cross-section through the badge.



Put ticks (✓) in the table to show which types of radiation reach each region of the photographic film. [3]

Radiation Region	Alpha	Beta	Gamma
A			
B			
C			

4. Complete the table below by choosing the correct word from each column. [5]

Radiation type	Ionising power Weak/moderate/strong	Charge Positive/none/negative	Mass Small/large/none	Penetration Low/moderate/high	Speed Slow/fast/very fast
ALPHA					
BETA					
GAMMA					

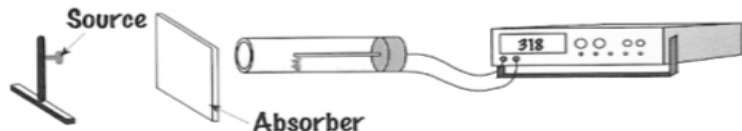
Radioactive Decay

1. What do we mean if we say a substance is radioactive?[1]

2. A teacher decided to do an experiment to find out which type(s) of radiation was emitted by a radioactive source.

- a) Initially a reading was taken with the radioactive source absent. The Geiger counter gave a reading of 24 counts per minute. What does this show?
-[2]

- b) The source was then introduced and the experiment was set up as shown



Different absorbers were placed between the source and the Geiger counter. The radiation reaching the Geiger counter was measured for one minute. The results are shown opposite:

Absorber Used	Average counts per minute
No absorber	1900
Card 1mm thick	1900
Aluminium	500
Lead 1cm thick	120

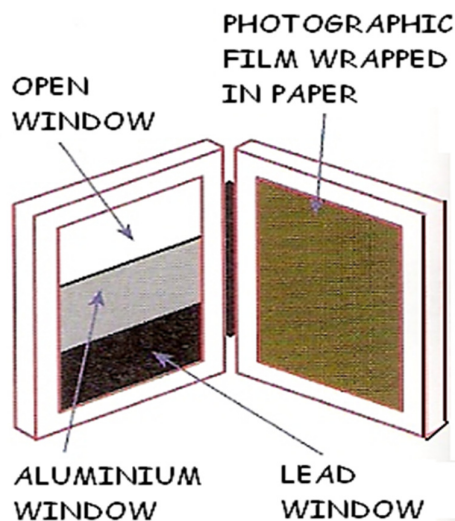
- i) What type or types of radiation did the source emit?[1]
 ii) Explain your answer

.....[2]

- iii) Why was the count rate reduced to 120 counts per minute with 1cm of lead but not down as low as background radiation

.....[2]

3. Nuclear power station workers wear a special film badge to check how much radiation they have been exposed to. A diagram of the badge is shown.



- a) Why is the photographic film wrapped in paper?

.....[1]

- b) What would happen to the photographic film on exposure to radiation?

..... [1]

- c) Explain how the film badge detects the different types of radiation

.....

.....

.....[1]

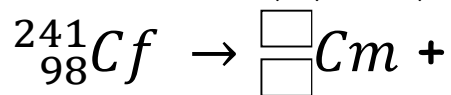
Nuclear Equations

1. Complete the table below:

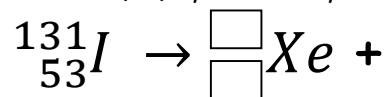
Radiation	Mass Number	Atomic Number	Charge	Symbol in equations
Alpha				
Beta				
Gamma				

2. The following radioisotopes decay by emission of radiation. Complete the equation for each: [7]

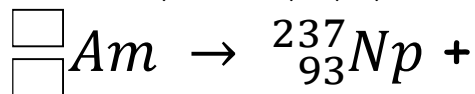
a) Californium (Cf) to Curium (Cm) by alpha decay



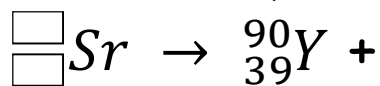
b) Iodine (I) to Xenon (Xe) by beta decay



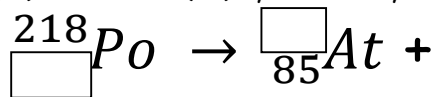
c) Americium (Am) to Neptunium (Np) by alpha decay



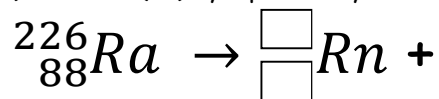
d) Strontium (Sr) to Yttrium (Y) by beta decay



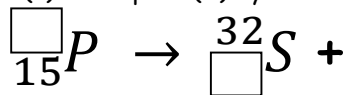
e) Polonium (Po) to Astatine (At) by beta decay



f) Radium (Ra) to Radon (Rn) by alpha decay



g) Phosphorus (P) to Sulphur (S) by beta decay



h) Which other radiation could also be emitted during the above decay processes?

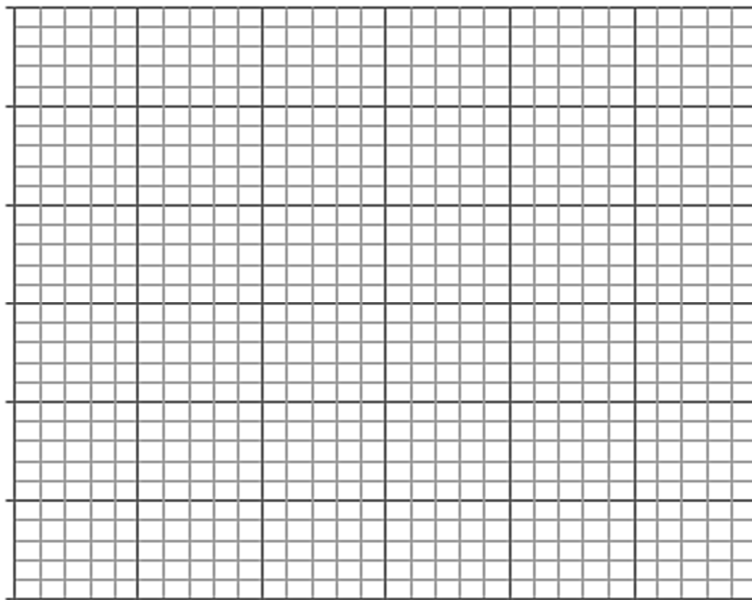
.....[1]

Half-life

- The of a sample always over time. Each time a decay happens, or is emitted. The half-life is the taken for of the radioactive to An isotope with a half-life decays more quickly than an isotope with a half-life. [11]
- An experiment was set up to calculate the half-life of a radioactive substance. Background radiation count rate was 30 counts per minute. With a radioisotope present the count rate was taken every five minutes for 20 minutes.

a) Complete the table [2]

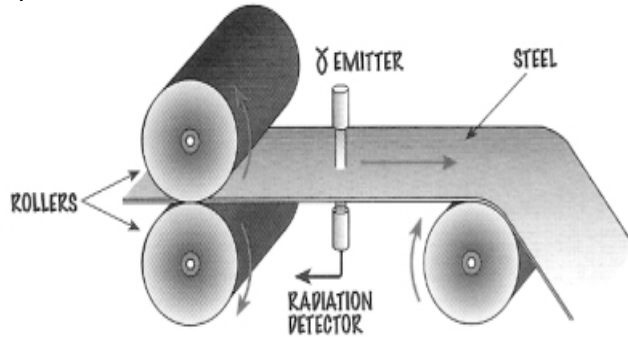
Time (min)	Count Rate (counts/min)	Corrected Count Rate
0	330	
5	180	
10	110	
15	75	
20	45	



- Plot a graph of time (x-axis) against corrected count rate (y-axis) [3]
 - Use the information to find the half life of the substance.....[1]
 - Use your graph to find the count rate after 17 minutes[1]
- Americium-241 is used in smoke detectors. It has a half-life of 460 years.
 - Why is it important for americium to have a long half-life?[1]
 - How long will it take americium to decay to one eighth of its original number of atoms?[2]
 - Iodine-138 is a radioactive substance. It has a half-life of 6s. A Geiger counter is used and the corrected count rate for a sample of iodine is found to be 2000 counts/min. How long will it take for the count rate to decrease to 125 counts/min?[2]
 - The relative amount of potassium-40 and argon in a sample of igneous rock can be used to date the rock. A rock sample contains 7 times as many argon atoms compared to potassium atoms
 - What fraction of the original potassium is left in the rock assuming there was no argon to begin with?[2]
 - The half-life of potassium-40 is 1300 million years. Calculate the age of the rock[2]

Uses of Radioactivity

1. The diagram shows a method of controlling the thickness of steel. A radioactive source which emits γ rays is placed on one side of the steel and the detector on the other side.



a) Explain how it works

.....

.....

.....

.....

b) Explain why a γ ray is used and not an α or β

.....

.....

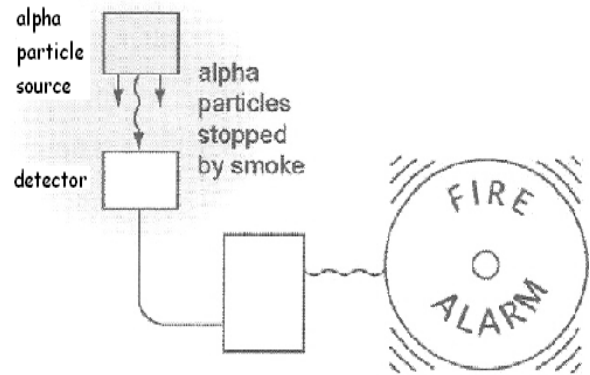
c) This setup can also be used to control the thickness of paper. What difference should be made? Explain

.....

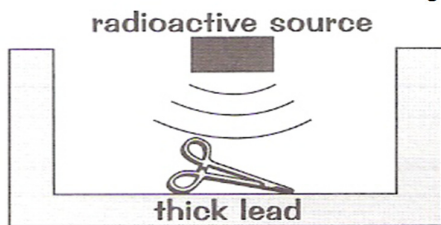
.....[2]

2. The following sentences explain how a smoke detector works, but they are in the wrong order.

	The circuit is broken so no current flows
1	The radioactive source emits alpha particles
	A current flows between the electrodes - alarm is off
	The alarm sounds
	The air between the electrodes is ionised by the alpha particles
	A fire starts and the smoke particles absorb the alpha radiation



3. Radiation can be used to sterilize surgical instruments.



a) What kind of radiation is used and why?

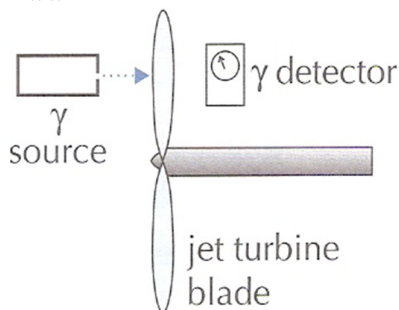
.....

.....[2]

b) What is the purpose of the thick lead?

.....[2]

4. Gamma radiation can be used to test turbine blades in jet engines.



a) Explain how the test would detect a crack in the blade

.....

.....

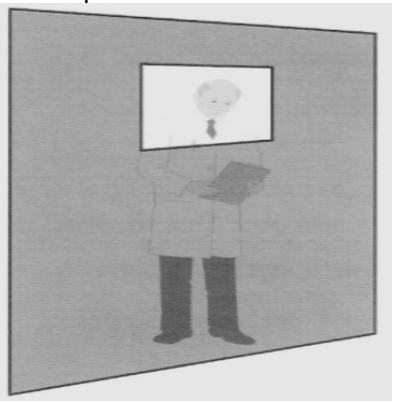
.....

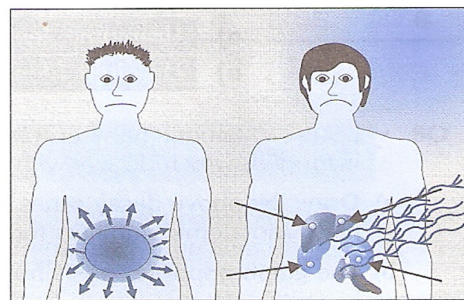
.....[2]

b) How is this method preferable to hitting the blades with a hammer to test their strength?

.....[2]

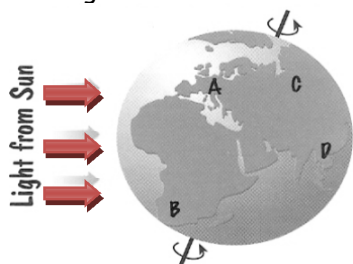
Dangers of Radioactivity

1. Radiation emitted from a radioactive material is often called 'ionising radiation'
 - a) Explain what is meant by the word 'ionising'[2]
 - b) Why is ionising dangerous to humans?
.....[2]
2. Gamma rays are emitted by radioactive substances. Medically they can be helpful or harmful to humans
 - a) Explain how gamma rays can be harmful to humans[2]
 - b) Explain how gamma rays can be helpful to humans[2]
3. The picture shows a worker at a nuclear plant
 
 - a) Name a suitable material for the protective screen[1]
 - b) How does this help protect him from radiation?[1]
 - c) If the screen was not there, what effect would each type of radiation have on him?
 - α[1]
 - β[1]
 - γ[1]
 - d) It is very important that workers do not ingest (eat or breath in) radioactive substances. What precautions can be taken to prevent this?
.....[2]
4. What type(s) of radiation are most dangerous OUTSIDE the body? Explain
.....[2]
5. What type(s) of radiation are most dangerous INSIDE the body? Explain
.....[2]
6. One of the scientists is taking the necessary precautions, while the other isn't. Describe 3 things which the careless scientist is doing wrong:
 - i.[3]
 - ii.
 - iii.



The Earth and the Solar System

1. The drawing shows the Earth:



a) State which place (A, B, C or D) is:

i) Experiencing Daytime[1]

ii) Experiencing night time[1]

iii) Experiencing Summer[1]

iv) Experiencing Winter[1]

b) Explain your answers:

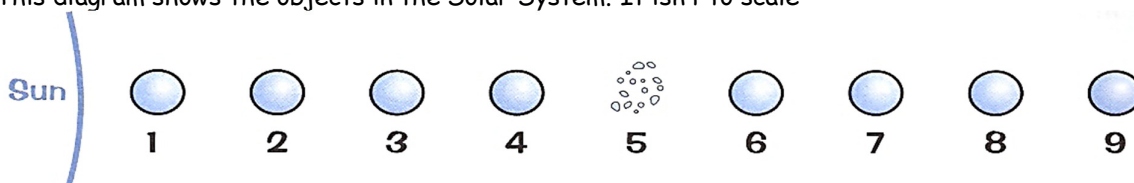
.....
[4]

2. Explain the different seasons using diagrams:

.....

[4]

3. This diagram shows the objects in the Solar System. It isn't to scale

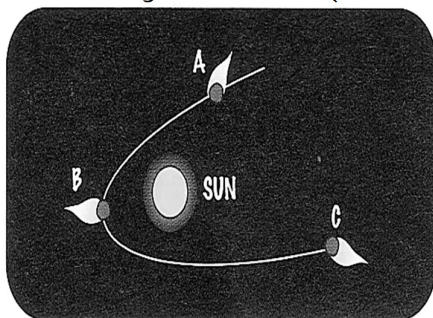


In the table below write the correct number under each object's name to show its position in the Solar System

Object	Mars	Earth	Asteroids	Venus	Saturn	Neptune	Jupiter	Mercury	Uranus
Number									

[10]

4. The diagram shows the path of a comet around the Sun:



a) Explain why the comet has a tail

.....[1]

b) Why does the tail always point away from the sun?

.....[1]

c) Explain why comets are not seen very often

.....[1]

d) What is the difference between asteroids and meteors?

.....[1]

5. There are eight planets in our solar system.

a) Name two planets that have a shorter orbit around the Sun compared to the Earth

i) ii)

b) Why is their period of orbit less than that of the Earth?

.....[1]

Satellites

1. Some satellites are put in orbit above the Earth at a height of 35600km and a speed of 3km/s. This gives them an orbit of 24 hours.

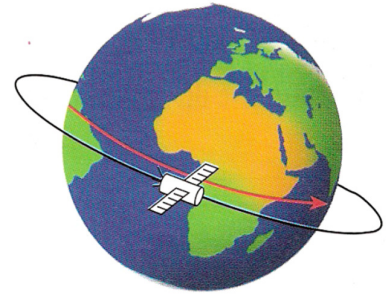
a) What is the name of this kind of orbit?

.....[1]

b) Suggest 2 suitable uses for this kind of orbit. Explain why it is suitable for each case

.....

.....[2]



2. The diagram shows a satellite that orbits the Earth once every ninety minutes

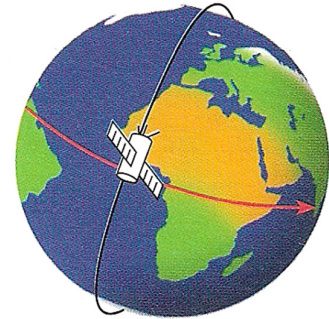
a) What is the name of this kind of orbit?

.....[1]

b) Suggest 2 suitable uses for this kind of orbit. Explain why it is suitable for each case

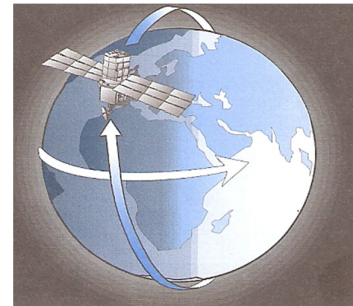
.....

.....[2]



3. Artificial satellites have been used for different purposes since the first successful launch in the 1950's. Today, satellites play an important role in our lives. The following can describe the motion of satellites:

A. A high orbit
B. A low orbit
C. Geosynchronous
D. Above the atmosphere
E. In a polar orbit
F. In an equatorial orbit
G. Orbits in a few hours
H. Orbits in 24 hours



Which of the statements above will apply to: (give reasons for your answers)

a) Communications satellites

.....

.....[2]

b) Most weather satellites

.....

.....[2]

c) Spy satellites

.....

.....[2]

d) Satellites broadcasting TV channels

.....

.....[2]

Gravitational Forces

1. The table below shows data about the orbits of six planets which orbit the sun

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn
Distance from the sun (A.U.)	0.39	0.72	1.00	1.52	5.20	9.54
Time for one orbit (years)	0.24	0.62	1.00	1.88	11.9	29.5

- a) What can you conclude from the data? Circle the letters of any of the following which apply:
- A. The further out a planet is from the Sun, the weaker the Sun's gravity
 - B. The time it takes a planet to orbit the Sun is directly proportional to the distance from the Sun
 - C. The further out the planet is from the Sun, the longer it takes to orbit
 - D. Planets further out than Saturn will take longer than 29.5 years to orbit the Sun

- b) Why are the orbits of geostationary satellites higher up than those of polar satellites?

.....
[2]

2. The force of gravity between two objects depends on their masses and the distance between them.

- a) If you double the distance you are from a planet, what happens to the force of its gravity on you?

.....[2]

- b) A satellite orbiting the Earth feels a gravitational force of 250N

What gravitational force would the same satellite feel if its orbit were moved to be:

- i) Five times further away from the Earth?
- ii) Half the distance from the Earth?

3. The diagram shows two planets. They both have the same mass M .



- a) Mark on the diagram the directions of the pull force acting on each planet

- b) Planet Z is twice as far away from X compared to planet Y and in the opposite direction. If planet X exerts a pull force of value F on planet Y, what force will it exert on planet Z? Explain your answer.

.....
[2]

- c) Two other planets A and B are the same distance apart as planets X and Y but each has a mass of $4M$. Describe and explain how the force between A and B will be different to the force between X and Y.

.....
[2]

- d) Two other planets P and Q have a mass of $8M$ and $2M$ respectively. At what distance apart, compared to X and Y do they need to be to have a force between them of F ?

.....
[2]

Stars

1. Describe how star like our Sun is formed

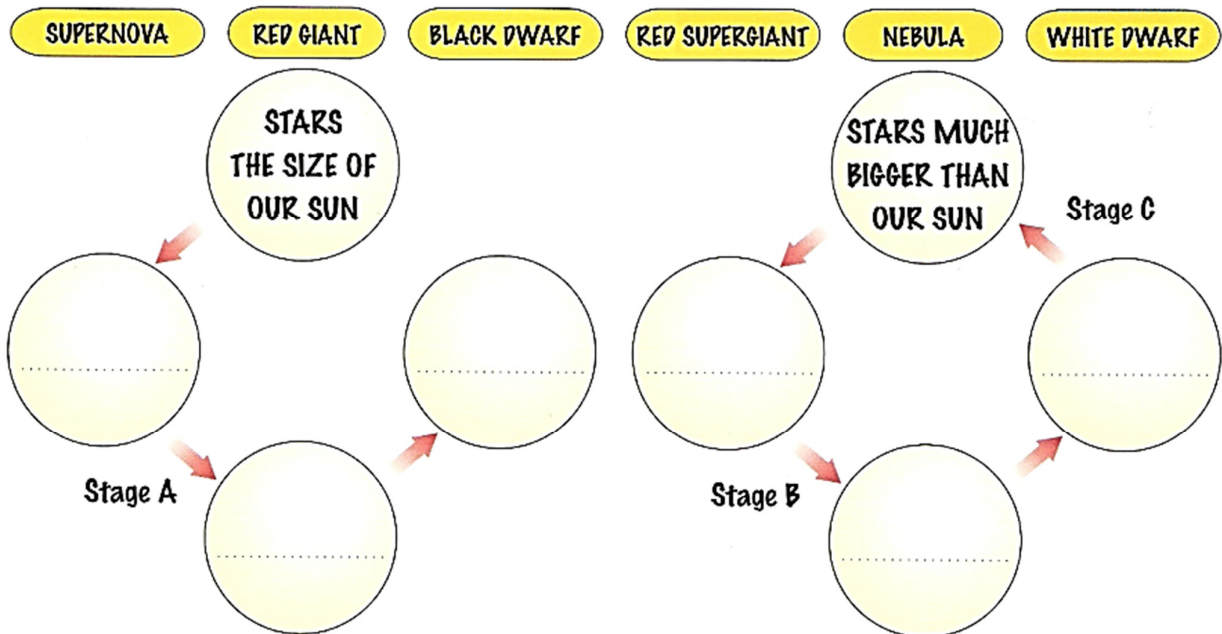
.....
[2]

2. Describe how the planets are formed

.....
[2]

3. The two flow diagrams show the cycle of change which occurs when a star DIES. Each circle shows what is formed at the each change.

- a) Complete each circle in the cycle by using the following words:



- b) Explain what effect gravity has on stage A

.....[2]

- c) Explain what happens at stage B

.....[2]

- d) Explain what happens at stage C

.....[2]

The Universe

1. A galaxy is a collection of billions of stars. Distant galaxies are known to be moving away from us and each other. The red shift of the light from a distant galaxy is a measure of the speed at which the galaxy is moving away from us. Hubble discovered that the further away a galaxy is, the greater is its red shift. This discovery led to the theory that the Universe is still expanding.

a) What is this theory called?[1]

b) What is meant by the term red shift in connection with light from a distant galaxy?

.....
[2]

2. Write down the correct words to complete the sentences below

a) When an ambulance is approaching you, the sound waves created by the siren are

This means that the sound has a pitch. After the ambulance has passed and it is travelling away from you, the sound waves from the siren are which results in a sound with a pitch. [4]

b) What is the name of the phenomenon

.....[2]

c) An astronomer studies a star which is known to be moving away from Earth. How does the motion of the star affect the light that she measures? Why is this?

.....
[2]

d) Decide whether the statements below are true or false:

- Distant galaxies appear to be moving away from us more slowly than nearby galaxies
- The universe appears to be expanding
- The Big Bang theory states that at one point in the past, all the matter and energy of the universe was concentrated into a very small space.
- Scientists know that the universe will keep expanding forever.

3. What evidence is there to support the idea that the Universe began with a 'Big Bang'? (Include a description of the RED SHIFT)

.....

[2]